Rabeya=43

ch=4

Polymerization

- Step Polymerization
- Addition Polymerization
  - Free Radical Polymerization
  - Ionic Polymerization
    - A. Cationic Polymerization
    - B. Anionic Polymerization

Q Define Step growth (condensation) polymerization with suitable example (1.5)

Step Growth (Condensation) polymerization

Tu-18

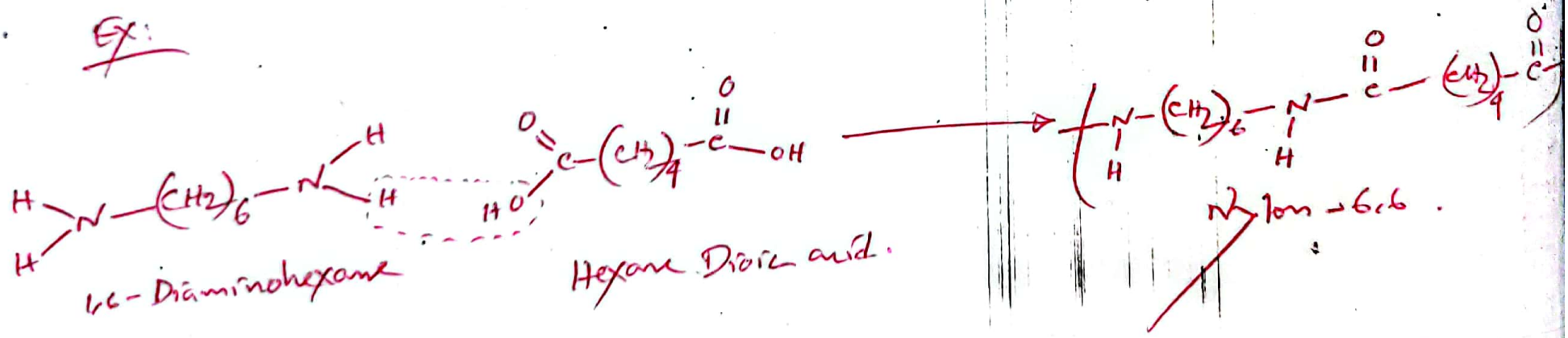
A condensation takes place between two bilyfunctional molecules to produce one larger polyfunctional molecule with the *possible* elimination of a small molecule such as H_2O, NH_3, HCl, etc. The reaction continues until one of the reagents is used up.

Common types:

- Polyamides
- Polyesters
- Polycarbonates
- Polyurethanes

example (Tu-20)

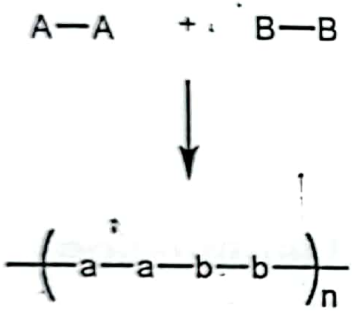
Ex:



Step Growth Polymerizations

Kabir
Che-46
AB

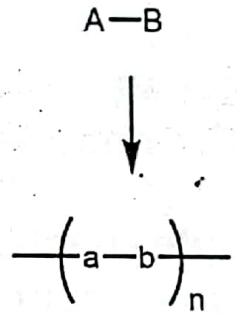
Bifunctional monomers



A-A + B-B ⟶ A-AB-B ⟶ B-BA-AB-BA-AB-BA-AB-B

1-mer 1-mer 2-mer poly-mer

AB monomers (Self condensation)



A-B A-BA-B $[A\text{-}B]_n$

1-mer 2-mer poly-mer

1. Difunctional monomers ⟹ Linear polymers

   AB type, AA and BB

2. Polyfunctional monomers ⟹ Network polymers

   Functionality > 2

3. Polymers retain their functionality as end groups at the completion of polymerization

4. Single reaction (cf. initiation, propagation, and termination in chain-reaction polymerization)

5. M.W. increases slowly

   Carothers equation

$$\overline{DP} = \frac{1}{1-p}$$

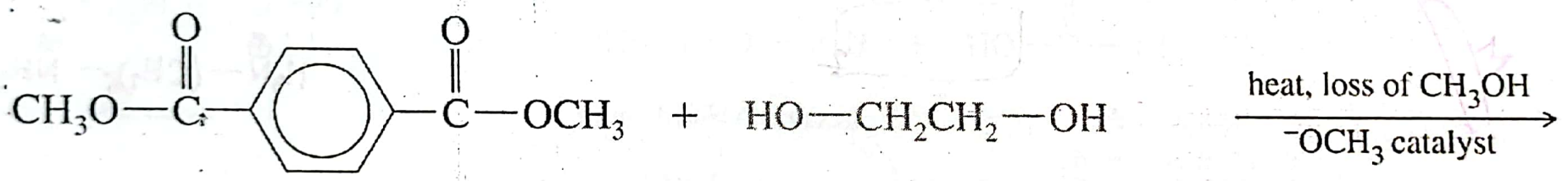
p = extent of reaction

6. High-yield reactions and an exact stoichiometric balance are necessary to obtain high-m-w linear polymer

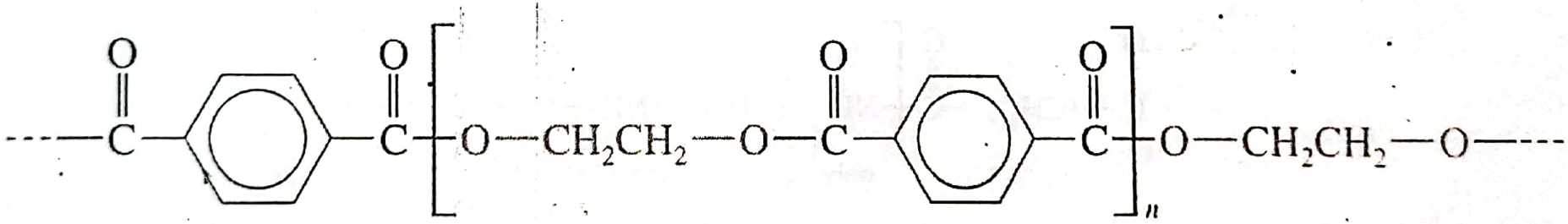
Kabir che-46

Polyesters (preparation by name প্রস্তুত 2টা)

- Polymer of terephthalic acid and ethylene glycol.
- Made by the transesterification of the methyl ester.



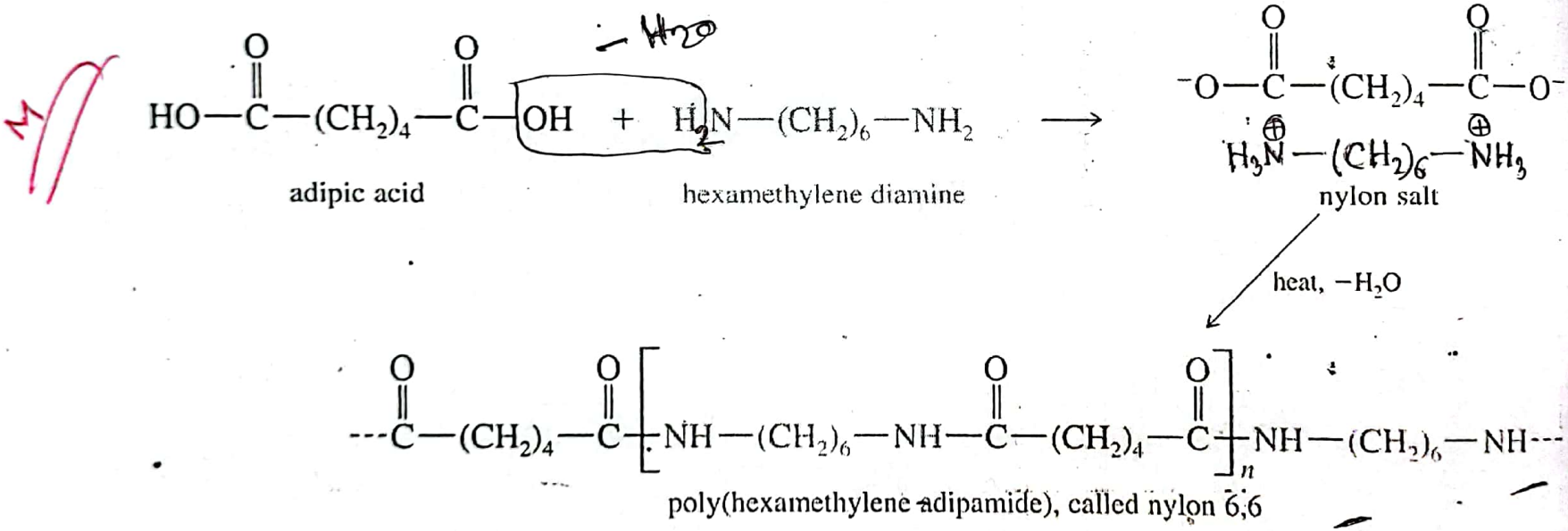
dimethyl terephthalate ethylene glycol



poly(ethylene terephthalate) or PET, also called Dacron® polyester or Mylar® film

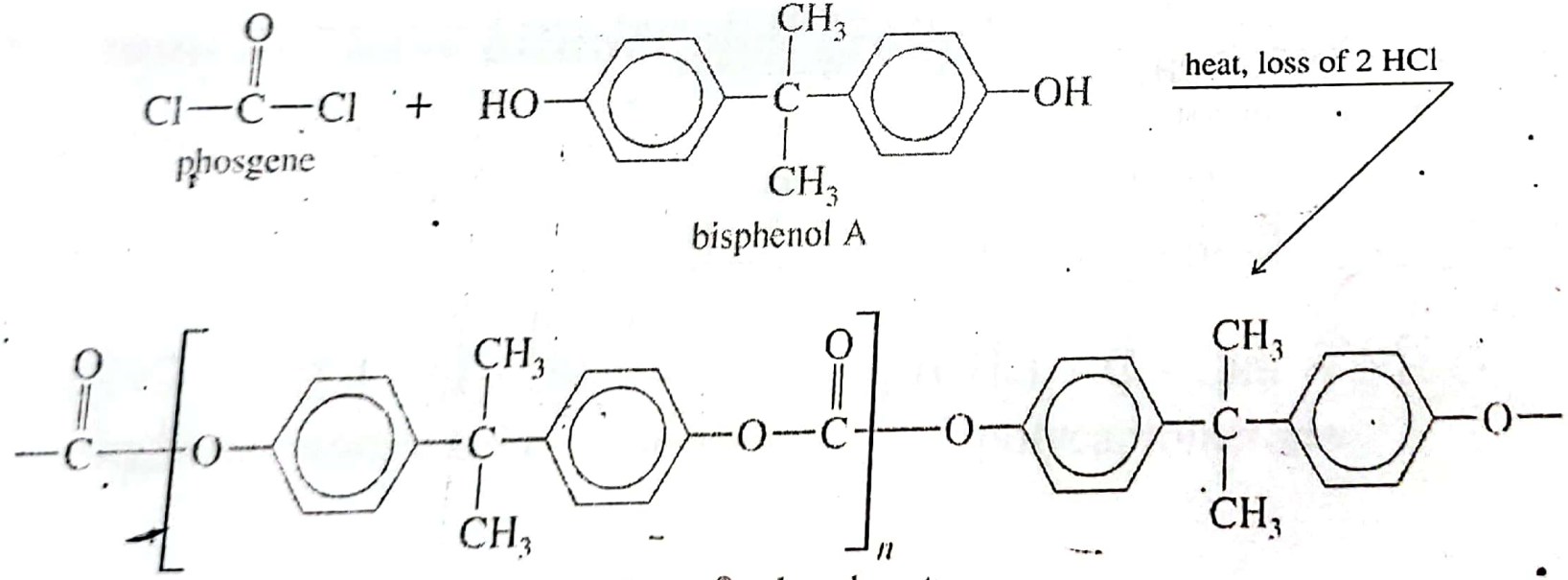
Polyamides: Nylon

Usually made from reaction of diacids with diamines, but may also be made from a single monomer with an amino group at one end and acid group at other.



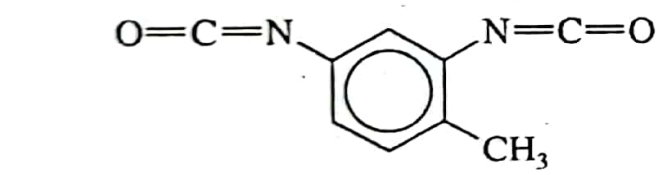
Polycarbonates

- Esters of carbonic acid.
- React phosgene with bisphenol A to obtain Lexan for bulletproof windows.



Polyurethanes

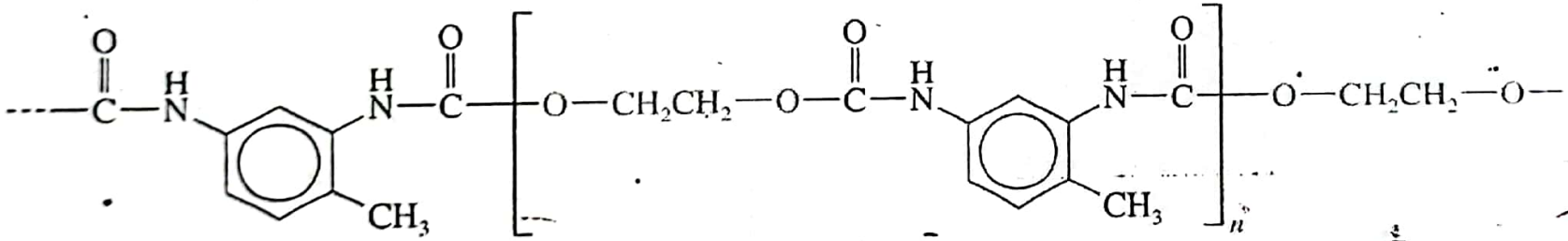
- Esters of carbamic acid, R-NH-COOH.
- Urethanes are prepared by reacting an alcohol with isocyanate.
- Polyurethanes are prepared by reacting a diol with a diisocyanate.



toluene diisocyanate

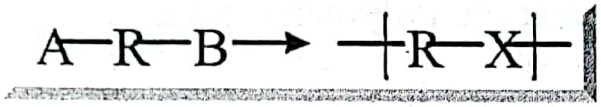
+ HO—CH_2CH_2—OH

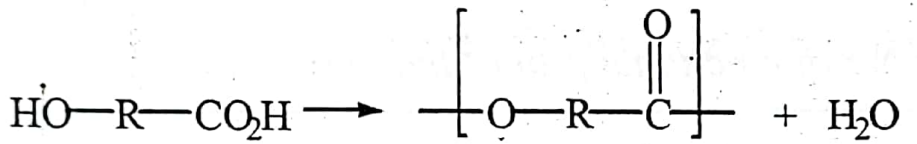
ethylene glycol

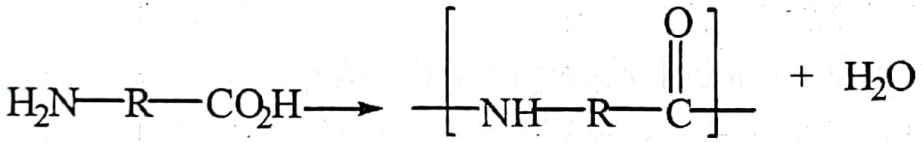


a polyurethane

One having both reactive functional groups in one molecule

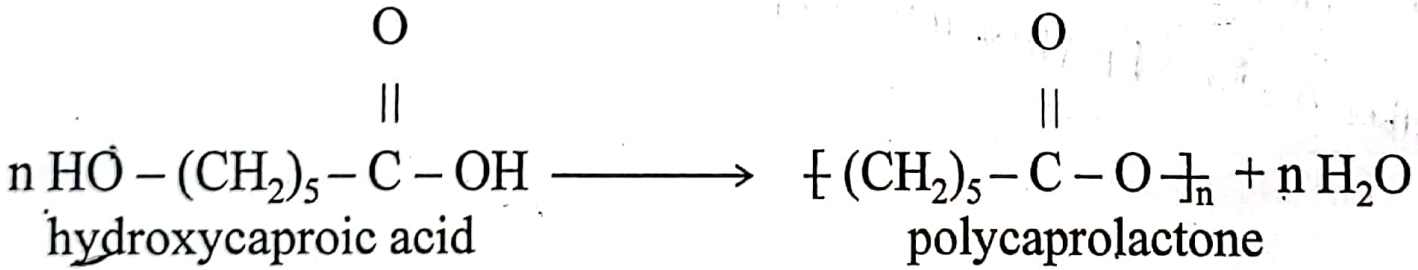






Monomer to have difunctional group

Self condensation



MW in Step-Growth polymerization

Polymer MW in step-growth polymerization is determined by the fractional conversion p, of the monomer during the polymerization.

Extent of reaction *p*:

$$p = \frac{\textit{number of functional groups that have reacted}}{\textit{number of functional groups originally present}}$$

One way to express molecular weight is the *degree of polymerization*- number of repeating units in the chain. Usually *average degree of polymerization* ($\overline{X_n}$ *or* $\overline{DP}$) is u as polymerization mechanism yields distribution of molecular weight.

MW কিভাবে express করা যায়?
Carother ও Flory DP দিয়ে MW express করা যায়।

Carothers equation 2016 * *

Consider a flask of monomer ....

If there are N_o molecules in the flask at time = 0 and N remaining at time t,

then the $\overline{DP}$ at time t is the average degree of polymerization... must just be, N_0/N.

The amount reacted is N_0-N and we can define p as the "conversion" or fraction reacted then as

$$p = (N_o - N) / N_o \text{ or}$$

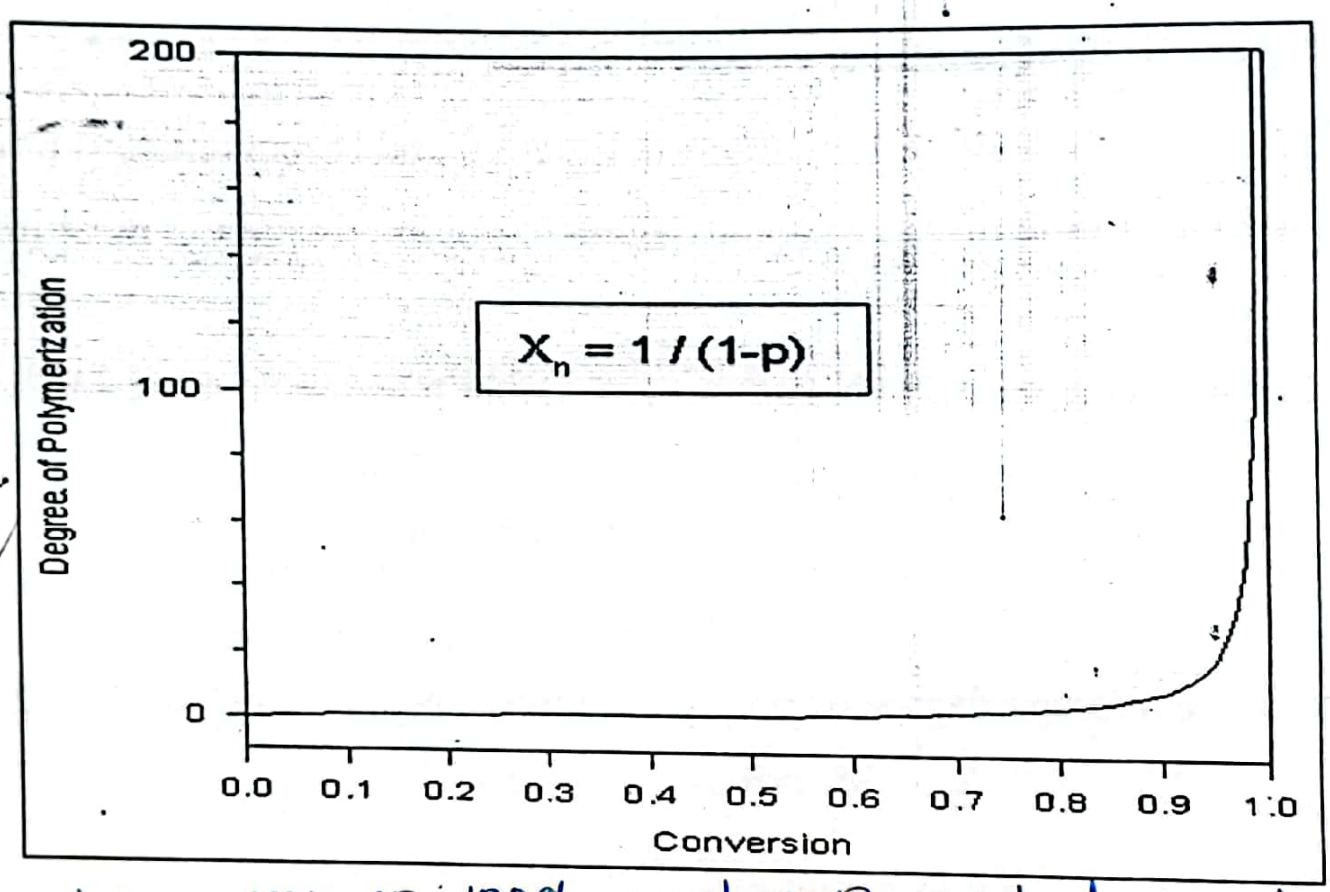
$$N = N_o(1 - P)$$

If $\overline{DP}$ is the average degree of polymerization... N_0/N ....substituting gives

$$\overline{DP} = 1/(1-P)$$

and for P = 0.98 (98% conversion), $\overline{DP}$ = only 50!

Carothers equation - relates/calculate molecular weight to the extent of a step growth polymerisation



Meaning of the graph: monomer 100% react করলে high degree of polymer পাওয়া যায়। initially ⇒ Dp কম। end stage ⇒ বেশি।

Number-average degree of polymerisation, $\overline{X}_n = N_0/N$

Conversion, $p = (N_0-N)/N_0 \longrightarrow N = N_0(1-p)$

Carothers equation

$$\overline{X}_n = 1/(1-p)$$

Number-average molecular weight: $\overline{M}_n = \overline{X}_n \times M_0 = M_0/(1-p)$
where M_0 is the molecular weight of the repeating unit

The weight-average degree of polymerisation, $\overline{X}_w = (1+p)/(1-p)$

Step – growth polymerization : MW distribution

Molecular weight distribution (polydispersity index) is $= \dfrac{\overline{X}_w}{\overline{X}_n} = \dfrac{(1+p)/(1-p)}{1/(1-p)} = (1+p)$

As $p \rightarrow 1$, MWD =2 for any linear step – growth polymerization .

MW-এর polydespersity মান বা ভিন্নতা ০1 MW মান. অসাম্য।

KINETICS OF STEP GROWTH POLYMERIZATION

WHY BOTHER ?

- How long does it take to make polymer ?
- Can we speed up the reaction ?
- What is the relationship between kinetics and the Mol. Wt. Of the product ?

8. Discuss the kinetics of step-reaction polymerization in the presence and absence of acid catalyst.

March 2016

KINETICS OF POLYCONDENSATION 2016

Let us consider the reaction below

$$\mathbf{A-A + B-B \longrightarrow A-AB-B}$$

Consider a polyesterification of bifunctionalmonomers in which initially equal amounts of di-basicacid and glycol are present.

$$HOOC-R_1-COOH + OH-R_2-OH \longrightarrow \text{-----} \overset{\overset{O}{\|}}{CO} \text{-----}$$

Case 1: Acid Catalyzed Polyesterification :

If the polyesterification is first order with respect to each functional group reactant, A and B, then the rate expression is

$$\text{Reaction Rate} = -\frac{d[A]}{dt} = k[A][B] \quad \text{--------(1)}$$

Given the fact that an exact stoichiometric balance is usually a necessary condition for achieving high molecular weight, we can equate [A] and [B] and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]^2 \quad \text{---------- (2)}$$

or, by integration,

$$\frac{1}{[A]} - \frac{1}{[A_0]} = kt \quad \text{----------(3)}$$

At any particular , t, $\overline{DP}$ is equal to the ratio of monomer molecules present initially to the total number at that time; that is

$$\overline{DP} = \frac{[A_0]}{[A]} \quad \text{---------- (4)}$$

Now the Carothers equation is

$$\overline{DP} = \frac{1}{1-p} \quad \text{---------- (5)}$$

Now combining eq. (4) and (5) and solving for [A],

$$[A] = [A_0](1- p) \text{ ---------- (6)}$$

By substituting this value into eq. (3), we obtain

$$\frac{1}{[A_0](1- p)} - \frac{1}{[A_0]} = kt \text{ ---------- (7)}$$

Since $\overline{DP} = 1/(1- p)$

$$\frac{\overline{DP}}{[A_0]} - \frac{1}{[A_0]} = kt \text{ ---------- (8)}$$

Or

$$\overline{DP} = [A_0]\, kt + 1$$

Case 2: No Acid Catalyst Added :

In the absence of added acid, the carboxylic acid assumes the role of catalyst and the rate expression becomes,

$$-\frac{d[A]}{dt} = k[A]\,[A][B]$$

That is

$$-\frac{d[A]}{dt} = k[A]^2[B] \quad \text{----------} \quad (1)$$

Again assuming [A] and [B] are equal,

$$-\frac{d[A]}{dt} = k[A]^3 \quad \text{----------} \quad (2)$$

Integration then gives

$$\frac{1}{[A]^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (3)$$

Substituting for [A] as above and rearranging

since, $[A] = [A]_0 (1-p)$

$$\frac{1}{[A_0]^2 (1-p)^2} - \frac{1}{[A_0]^2} = 2kt \quad \text{----------} \quad (4)$$

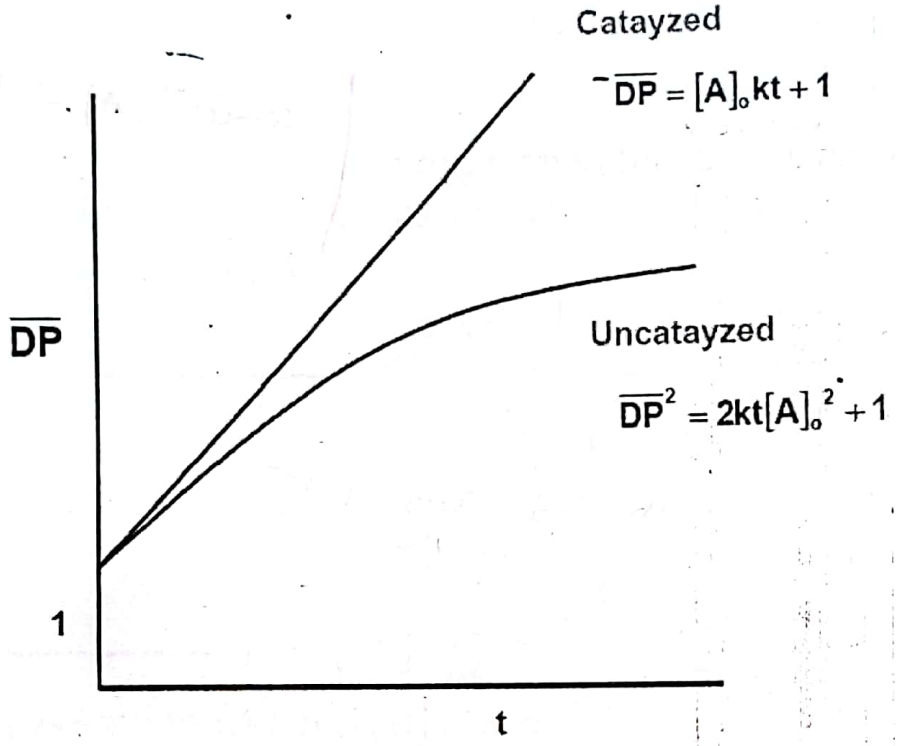
Or

Multiplying both side by $[A]_0^2$

$$\frac{1}{(1-p)^2} = 2kt\,[A_0]^2 + 1 \quad \text{----------} \quad (5)$$

and

$$\overline{(DP)^2} = 2kt\,[A_0]^2 + 1$$



The following conditions must be fulfilled in step-growth polymerizations for higher molecular weight:

- high conversions
- high monomer purities
- high reaction yields
- stoichiometric equivalence of functional groups

* Exact stochiometric balance is necessary to achieve high M.W.

balance হলে → high polymer M.W
imbalance " → low M.W · limit করার condition.

Stoichiometric Imbalance

Despite the fact that an exact stoichiometric balance is necessary to achieve high molecular weights in most instants of step reaction polymerization, in some important application it is necessary to limit the molecular weight.

•Epoxy resins are **polyethers** that are processed to an **oligomer stage** and are subsequently converted to **network polymer** by appropriate reactions of terminal epoxyide groups.

•With **polyimides** for fiber applications, molecular weight must often be limited because **too high a viscosity** is detrimental to **extrusion of filaments through the fine holes of a spinneret.**

M
2016 (C)

Three ways to **limit M. W.** in step polymerization **

1. **To quench the polymerization reaction by rapid cooling when the desired M.W. is attained.**
2. **To use an excess of one monomer.**
3. **To use small amounts of monofunctional reactant**

Stoichiometric imbalance

When the nonstoichiometric amount of functional groups is used, the relation between $\overline{DP}$ and the reaction conversion can be modified by modification of the Carothers equation.

In order to do so let us introduce a factor, *r,* representing the *stoichiometric imbalance.*

If the number of A and B functional groups present at the beginning of AA + BB polymerization is N_A^0 and N_B^0 respectively,

$$r = \frac{N_A^0}{N_B^0}$$

By convention r is always less than unity.

p = reaction conversion, in this case fraction of A group that have reacted.

As reaction of each A group consumes one B group, fraction of B reacted at p is equal

$$pN_A^0 \text{ or } prN_B^0$$

The number of *unreacted* groups , N_A and N_B is given by

$$N_A = (1-p)\,N_A^0 \quad \text{and} \quad N_B = (1-pr)\,N_B^0 = (1-pr)\,\frac{N_A^0}{r}$$

At this point, the number of A and B end group is equal to $N_A + N_B$, and because there are two end groups per molecule, the number of molecular chains, N, is given by

$$N = \frac{1}{2}(N_A + N_B)$$

That is

$$N = \frac{1}{2}\left[(1-p)\,N_A^0 + (1-pr)\frac{N_A^0}{r}\right]$$

which reduces to

$$N = \frac{N_A^0}{2}\left(1 + \frac{1}{r} - 2p\right)$$

We know that one repeating unit is formed every time an A and B functional group react; therefore the total number of repeating units N_r is given by

$$N_r = \frac{1}{2}(N_A^0 + N_B^0)$$

Since, $r = \frac{N_A^0}{N_B^0}$

$$N_r = \frac{1}{2}\left(N_A^0 + \frac{N_A^0}{r}\right) = \frac{N_A^0}{2}\left(\frac{r+1}{r}\right)$$

The average degree of polymerization is equal to the number of monomeric units divided the number of chains; that is

$$\overline{DP} = \frac{N_r}{N} = \frac{(N_A^0/2)[(r+1)/r]}{N_A^0/2[1+(1/r)-2p]}$$

$$r = \frac{N_A^0}{N_B^0} < 0, \quad (N_A^0 < N_B^0)$$

Carothers equation reduces to

$$\overline{DP} = \frac{r+1}{r+1-2pr}$$

Now if $r = 1$

$$\overline{DP} = \frac{1}{1-p}$$

When monomer A is completely used up in the polymerization (that is $p = 1$) the equation becomes

$$\overline{DP} = \frac{1+r}{1-r}$$

If monofunctional reagent is added

Imbalance factor

$$r' = \frac{N_A^0}{N_B^0 + 2\,N^0_{B'}}$$

where $N^0_{B'}$ = number of monofunctional B groups

Factor 2 : each B' molecule is equally as effective as one excess BB monomer in limiting the M.W.

Molecular weight distribution

In any polymerization involving difunctional monomer, each step reaction links two monomer together; thus the number of monomer units in the polymer is always one more than the functional group.

If a polymer contains x monomer units, the number of A or B groups that have reacted is x-1.

The probability that x-1 A or B groups has reacted is p^{x-1} , where p is the reaction conversion, defined previously as

$$p = \frac{N_0 - N}{N_0}$$

The probability of finding an unreacted group is

$$1 - p$$

The probability of finding a molecule containing x units and an unreacted A or B group is then

$$p^{x-1}(1-p)$$

If N= total number of molecule present at time t, the fraction that contains x units N_x, is given by

$$N_x = N\,p^{x-1}(1-p)$$

As $N/N_0 = 1-p$ (Carother equation), we can write the expression for N_x as

$$N_x = N_0(1-p)^2\,p^{x-1}$$

Where N_0 = initial number of monomers and the relationship defines number or mole fraction distribution at t.

The weight fraction w_x, of molecules containing x units is equal to the ratio of the mass of those molecules divided by the total mass; that is

$$w_x = \frac{xN_xM_0}{N_0M_0} = \frac{xN_x}{N_0}$$

Where M_0 is the mass of the repeating unit.

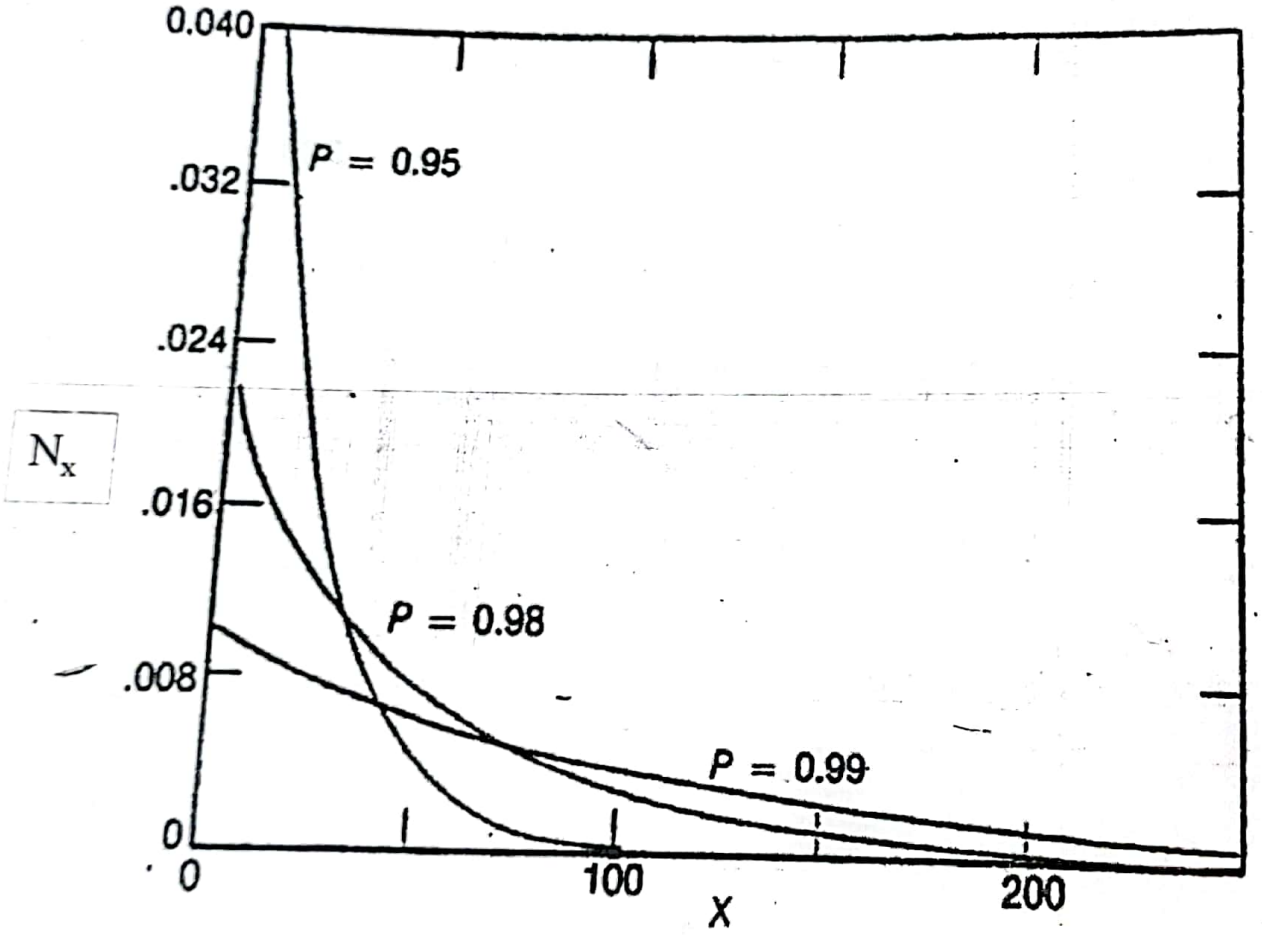
As, $N_x = N_0(1-p)^2 p^{x-1}$

We can write ,

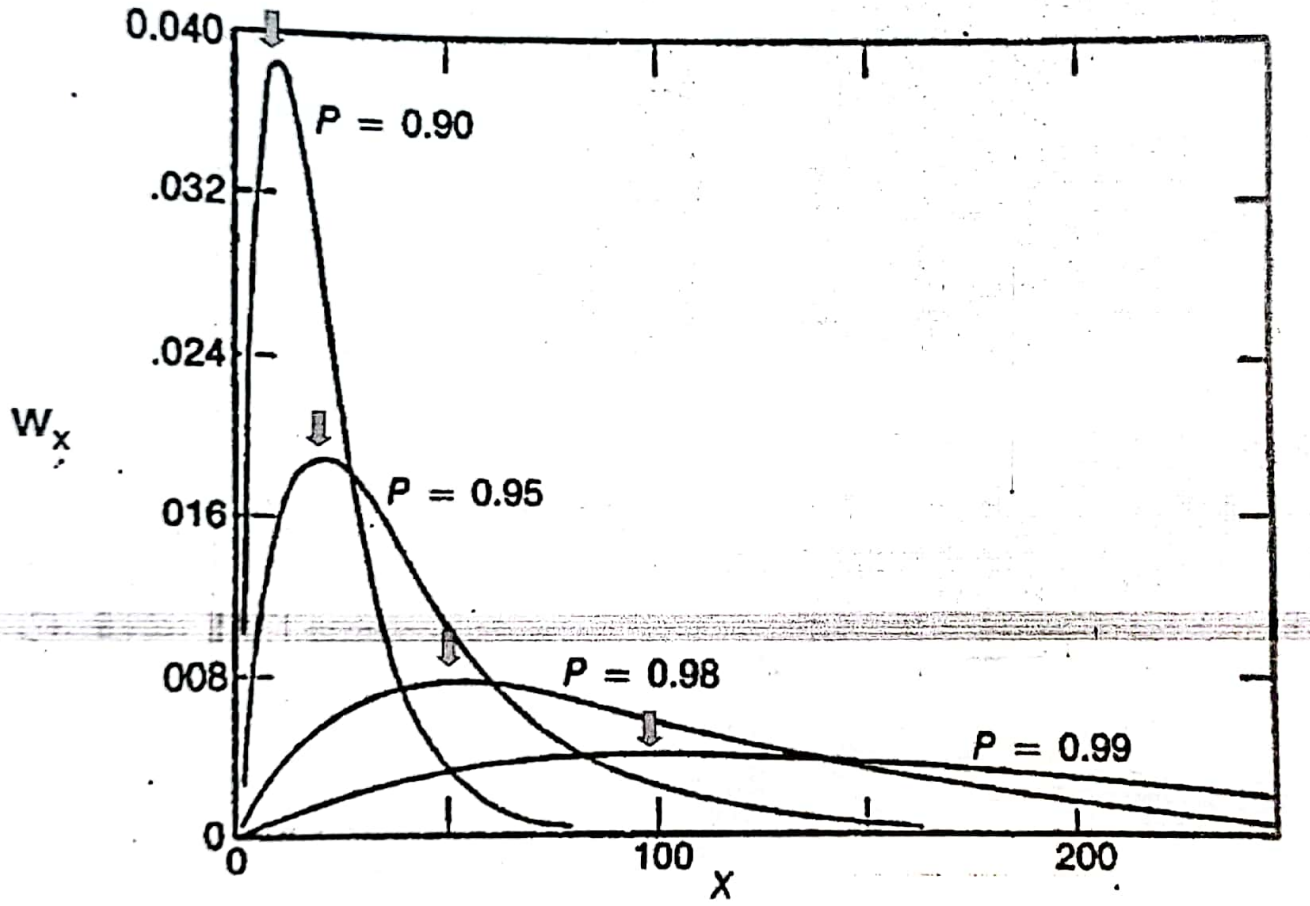
$$w_x = x(1-p)^2 p^{x-1}$$

Which is the weight fraction distribution.

Mole fraction distribution in linear step-reaction polymerization



: Weight fraction distribution in linear step-reaction polymerization



Addition Polymerization

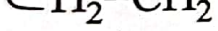
In addition polymerization (chain-growth polymerization), one monomer is added to the growing chain at a time.

m-mer + monomer = (m + 1)-mer

Characteristics
Free radical
Ionic: Anionic
Cationic
Living/Immortal

Examples

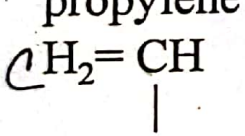
ethylene



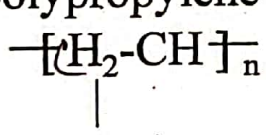
polyethylene



propylene



polypropylene



Just মনে রাখার জন্য

Characteristics of Chain-Growth Polymerization *

1. Only growth reaction adds repeating units one at a time to the chain
2. Monomer concentration decreases steadily throughout the reaction
3. High Molecular weight polymer is formed at once; polymer molecular weight changes little throughout the reaction.
4. Long reaction times give high yields but affect molecular weight little.
5. Reaction mixture contains only monomer, high polymer, and about 10^{-8} part of growing chains.

fundamental difference

Free Radical Polymerization

Addition polymerization can be initiated by free radicals. Free radicals are the molecules with unshared electrons. Well-known free radicals are produced by fragmentation of molecules such as benzoyl peroxide and azobisisobytylnitrile by homolytic scission (homolysis) of a single bond by heat.

Free Radical Polymerization Mechanisms

1. Overview – Free radical polymerization processes involve at least three mechanistic steps.

A. Initiation

1. Radical Formation (Generation)

$$In—In \xrightarrow[uv,\ \text{etc.}]{\Delta} In^{\bullet} + In^{\bullet}$$

2. Initiation

$$In^{\bullet} + M \longrightarrow In—M^{\bullet}$$

$$I \xrightarrow{kd} 2R^{\bullet}$$

Initiator (I) decomposes to yield two radicals

$$R^{\bullet} + M \xrightarrow{k_i} M_1^{\bullet}$$

Radical reacts with monomer to yield monomer-ended radical (M_1)

B. Propagation

$$In\text{-}M_1\cdot + M_2 \longrightarrow In\text{-}M_1M_2\cdot$$

$$In\text{-}M_1M_2\cdot + M_3 \longrightarrow In\text{-}M_1M_2M_3\cdot$$

$$In\text{-}M_1M_2M_3\ldots M_X\cdot + M_Y \longrightarrow In\text{-}M_1M_2M_3\ldots M_XM_Y\cdot$$

$$M_1^{\bullet} + M \longrightarrow M_2^{\bullet}$$

$$M_2^{\bullet} + M \longrightarrow M_3^{\bullet}$$

$$M_i^{\bullet} + M \xrightarrow{kp} M_{i+1}^{\bullet}$$

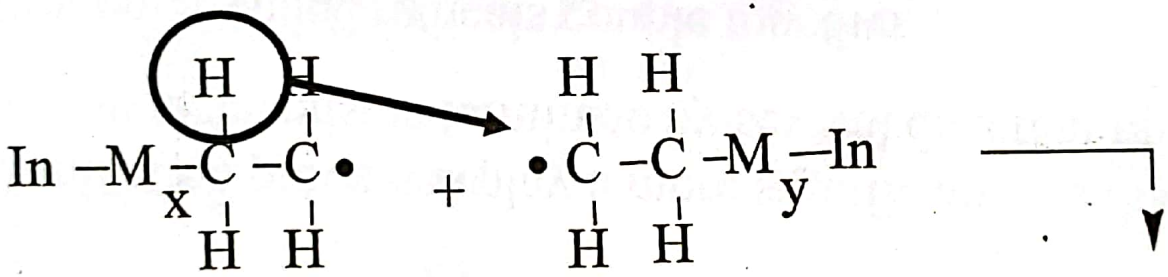
C. Termination

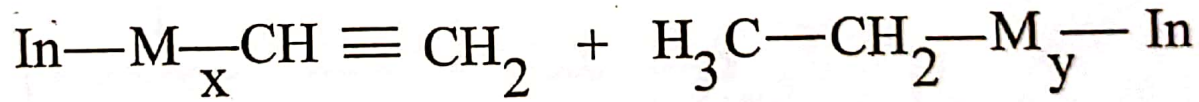
1) Radical Coupling (Combination)





2) Disproportionation (β-hydrogen transfer)





dead polymer chain

Chain Termination

In polymerization, termination processes play a more significant role than in small molecule reactions. Chain mechanisms terminate by several different processes…

Combination - two monomer-ended radicals couple together

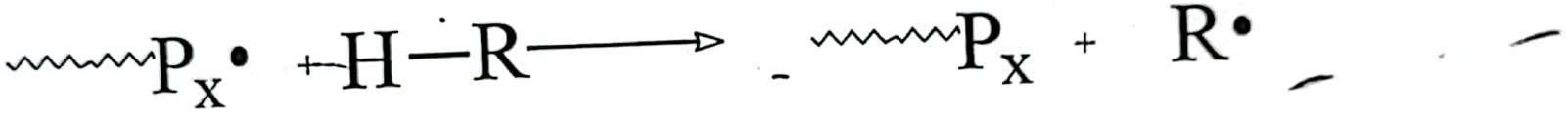
Disproportionation - chain end transfers a hydrogen to form two inactive molecules

Chain transfer - active end is transferred to another species (chain, monomer, initiator,solvent)

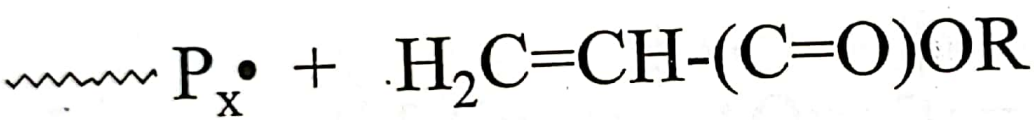
Reaction with impurities

D. Chain Transfer (sometimes) – An atom is transferred to the growing chain, terminating the chain growth and starting a new chain.

Chain Transfer to Chain Transfer Agent:



Chain Transfer to Monomer:



Chain Transfer to Polymer:

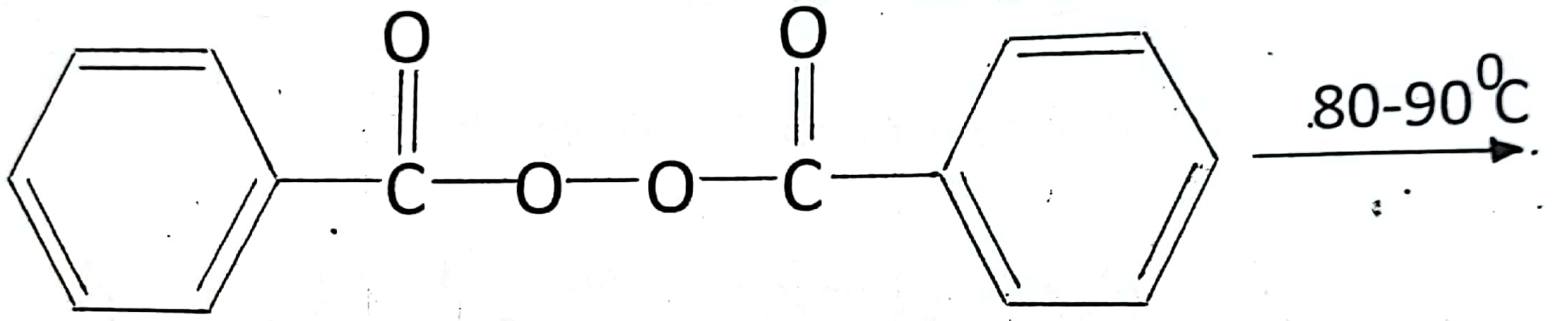
Causes Branching

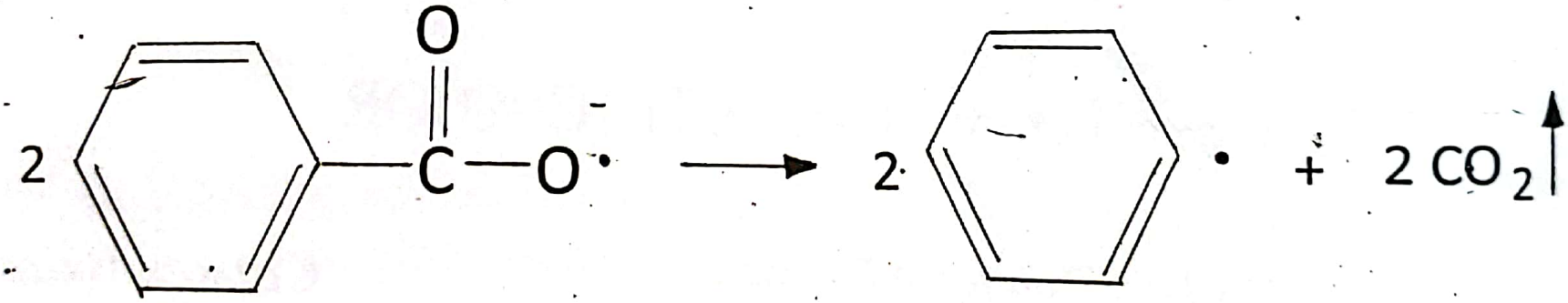


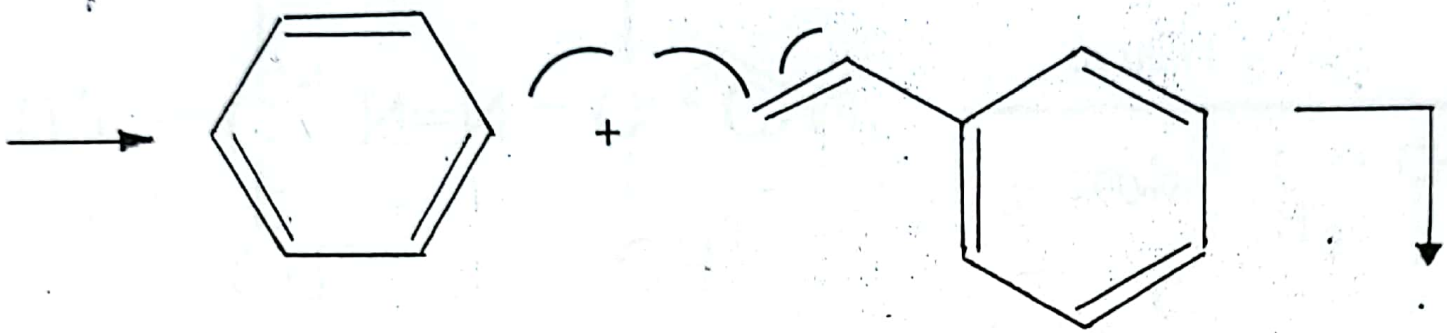
1. Initiation – "Getting the thing started!"

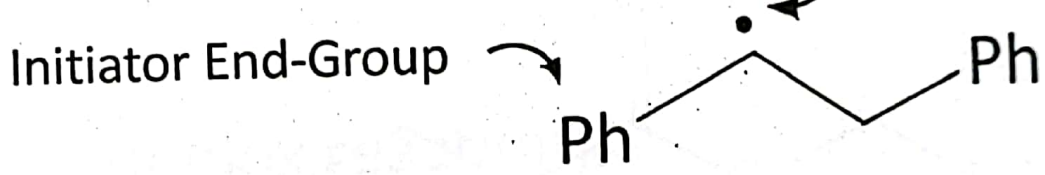
A. Radical Generators (Initiators)

1. Benzoyl Peroxide

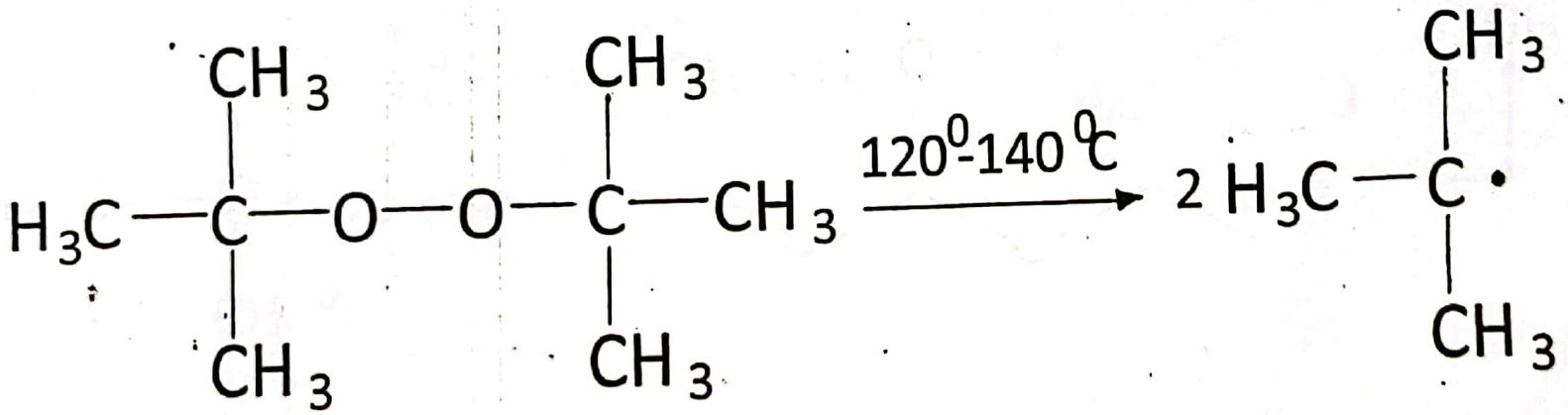


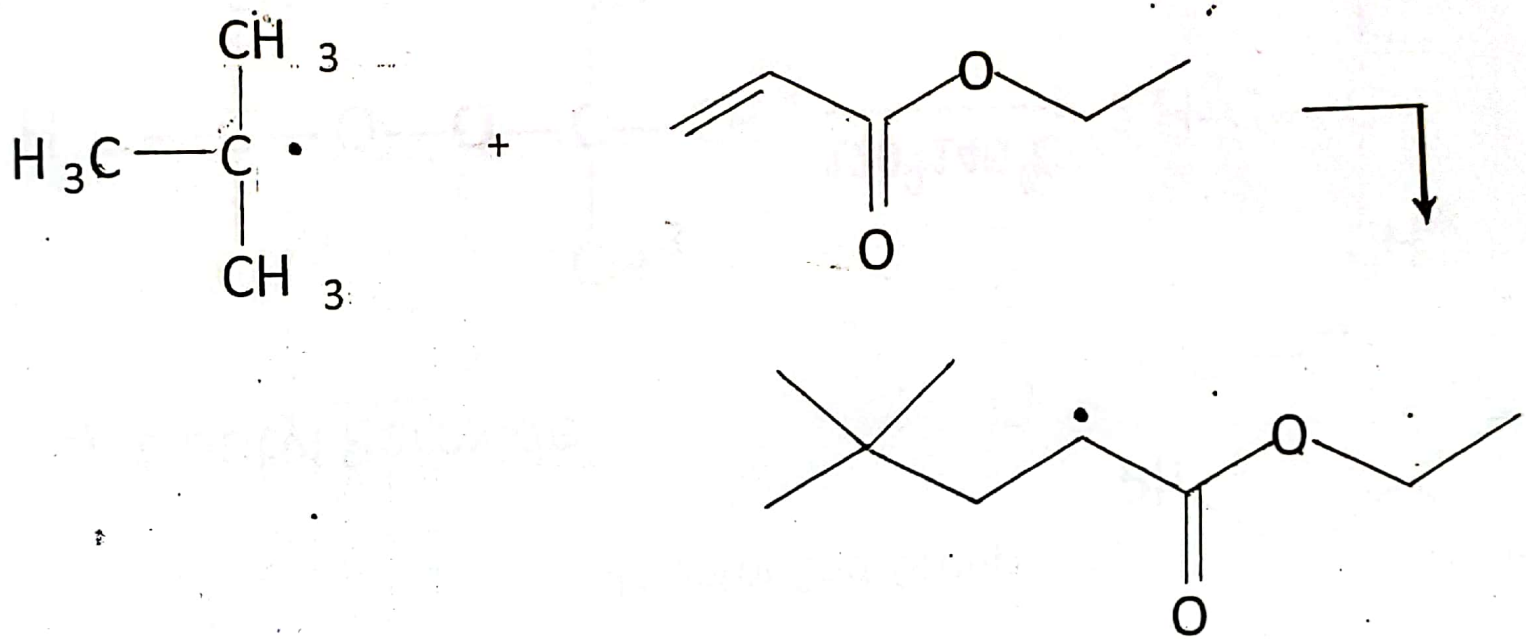




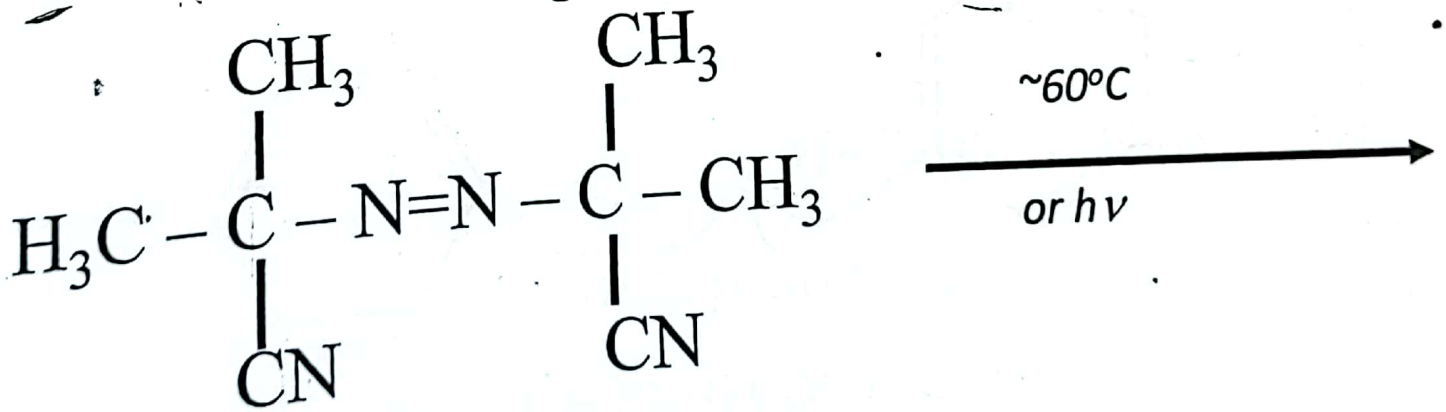


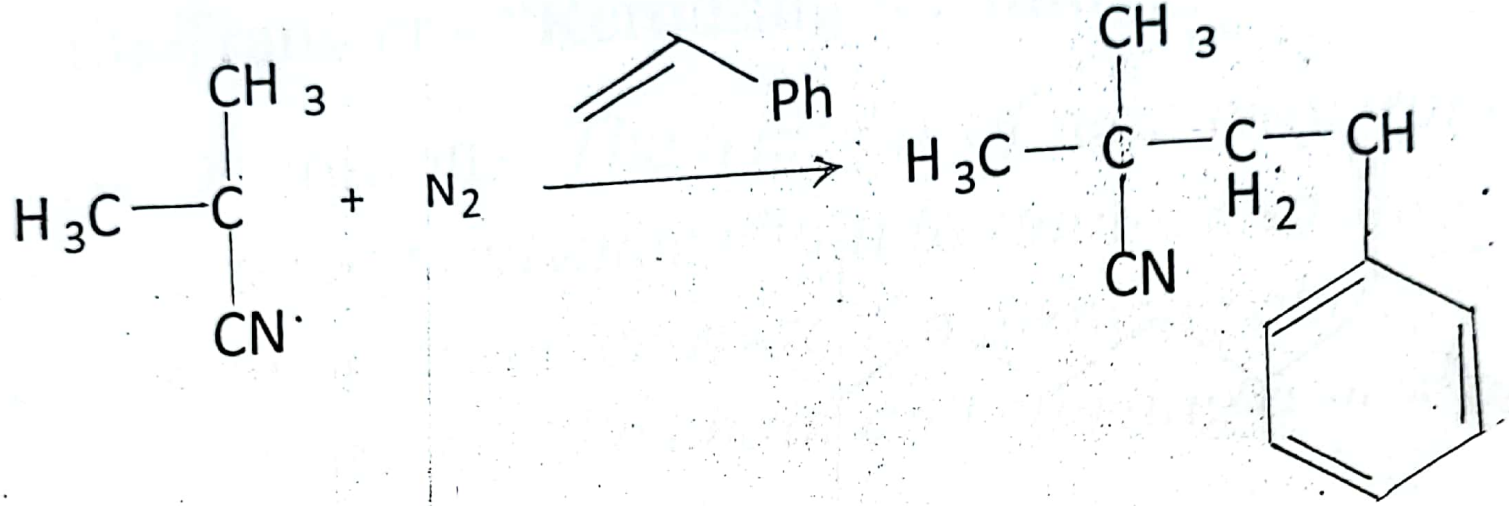
2) *t*-Butyl Peroxide



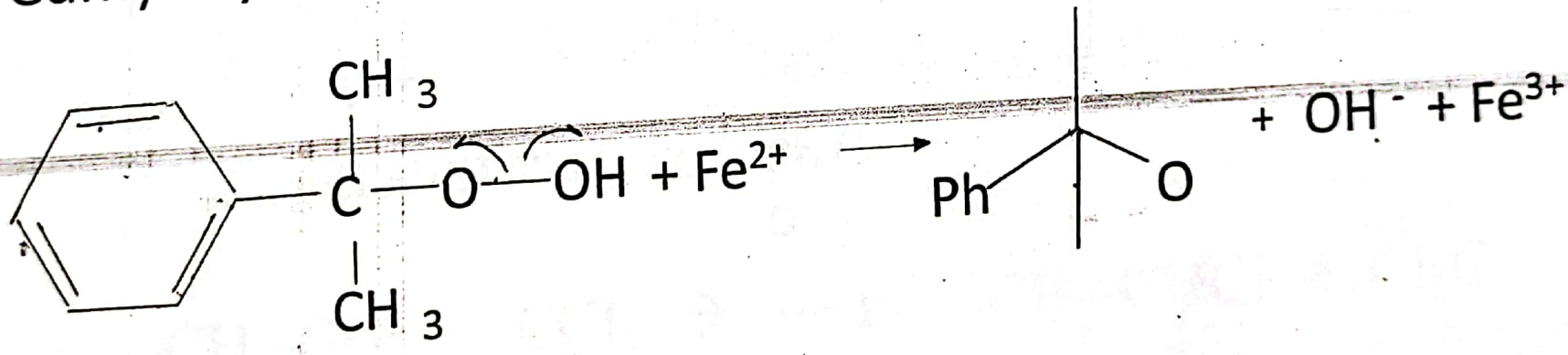


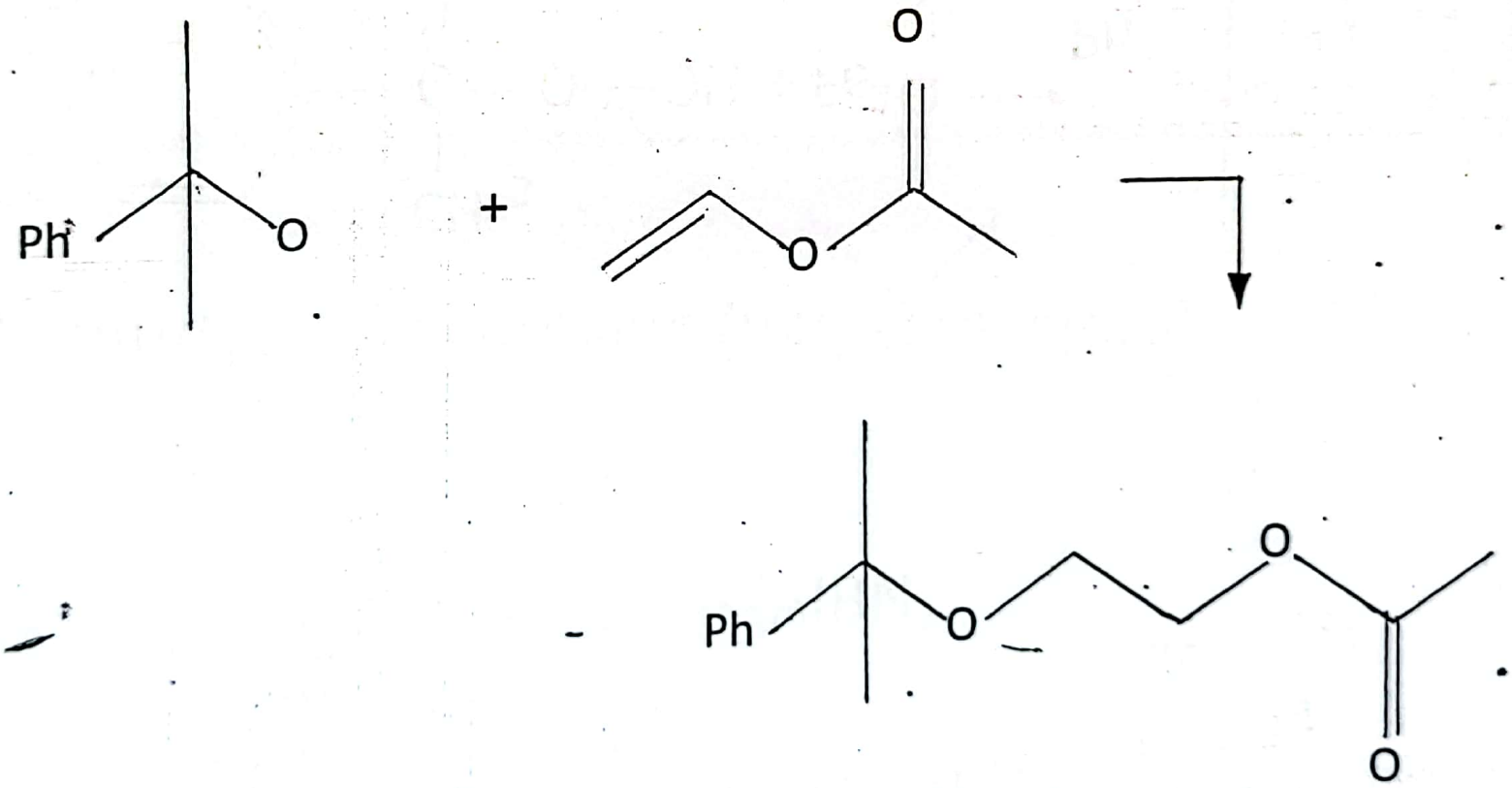
3) Azobisisobutyronitrile (AIBN)





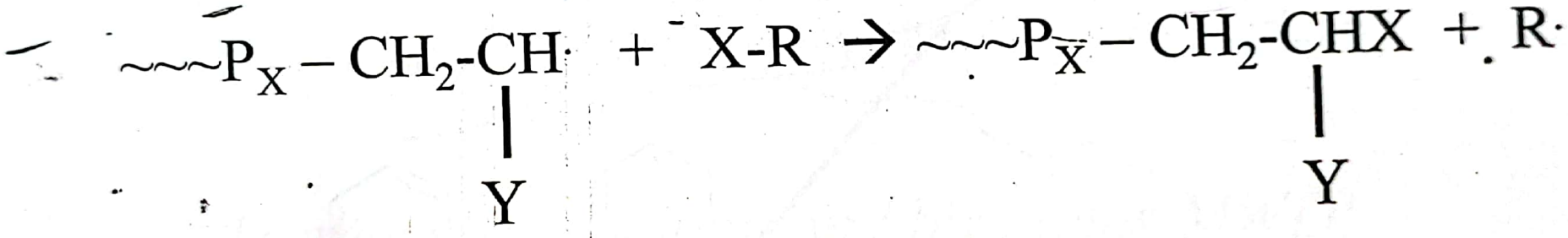
4) Cumyl Hydroperoxide (Redox reaction)



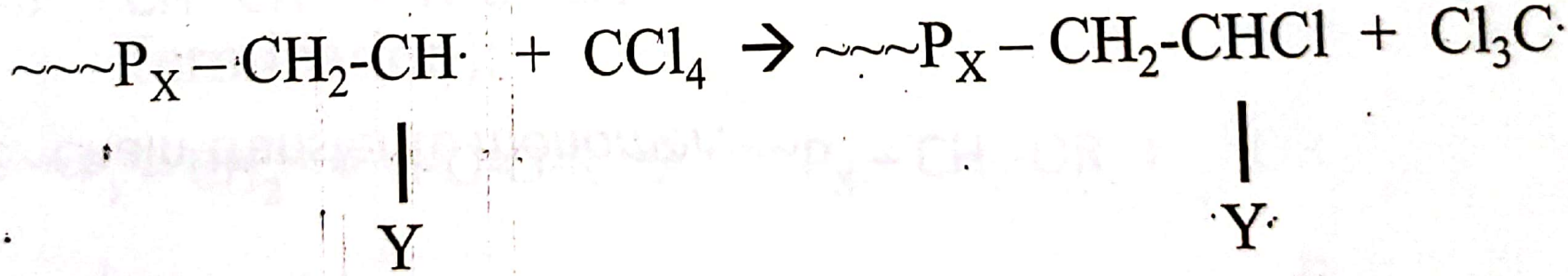

Ph
O
+
O
O
Ph
O
O
O

Chain-Transfer - "Rerouting the thing!"

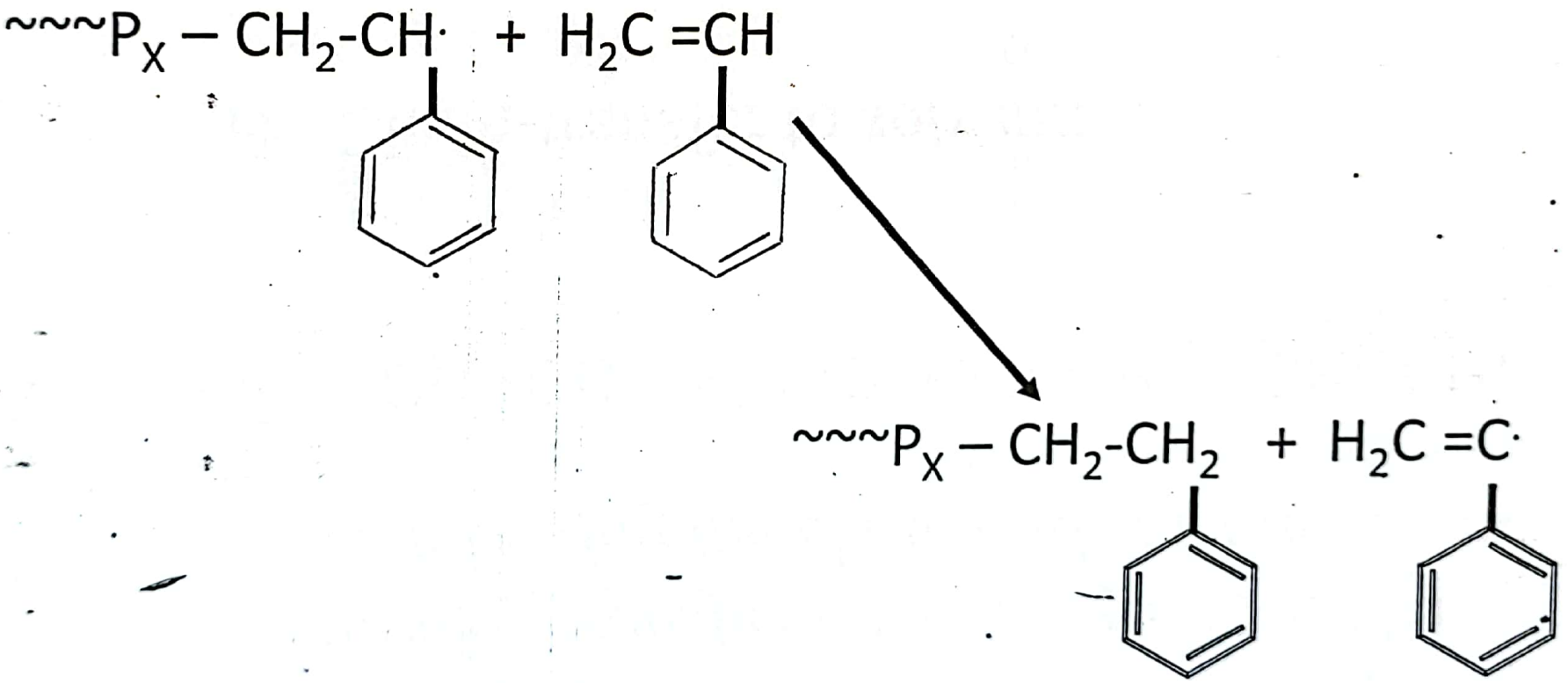
A. Definition – *The transfer of reactivity from the growing polymer chain to another species. An atom is transferred to the growing chain, terminating the chain and starting a new one.*



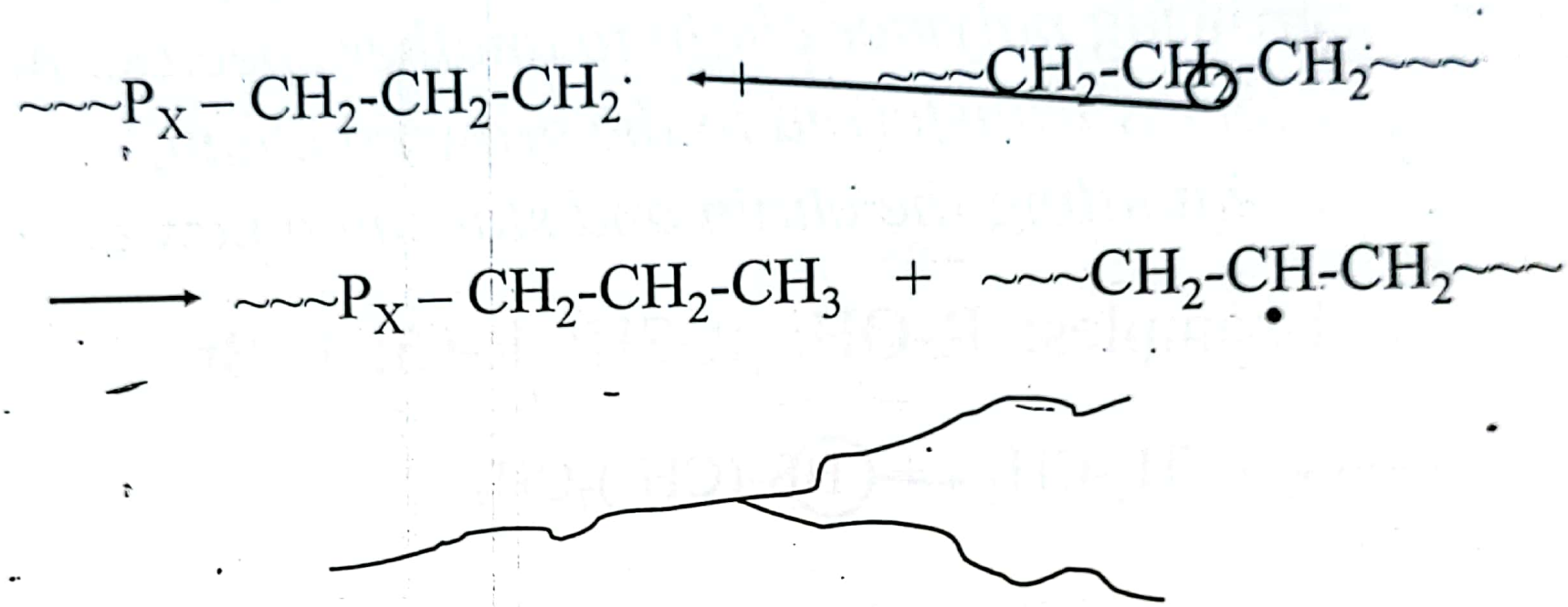
B. Chain-transfer to *solvent*:



C. Chain-transfer to *monomer*:



D. Chain-transfer to *polymer*:



Increases branching and broadens MWD!

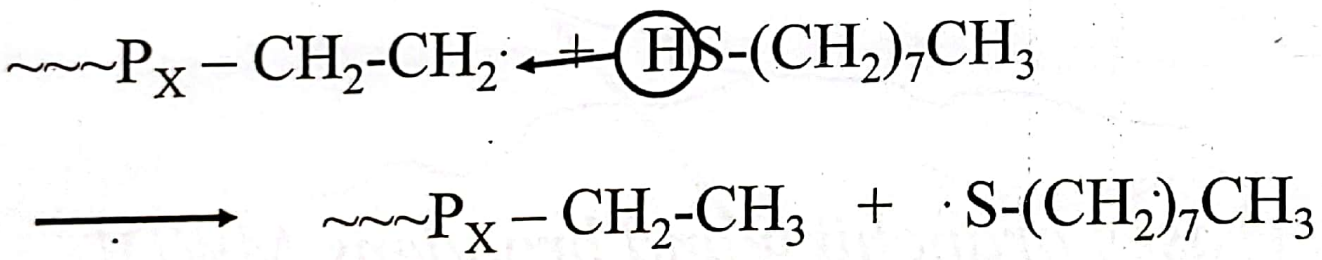
E. Chain-transfer to *Initiator* (Primary Radical Termination):

~~~P$_X$ – CH$_2$· + R-O-O-R → ~~~P$_X$ – CH$_2$-OR + ·OR
~~~

F. Chain-transfer to *Chain-transfer Agent:*

Definition – *The transfer of reactivity from the growing polymer chain to another species. An atom is transferred to the growing chain, terminating the chain and starting a new one.*

Examples: R-OH; R-SH; R-Cl; R-Br

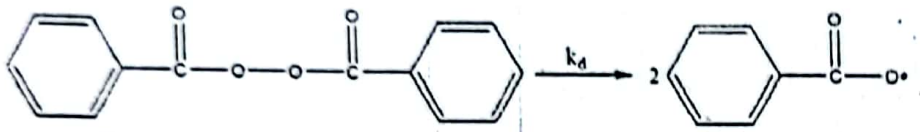


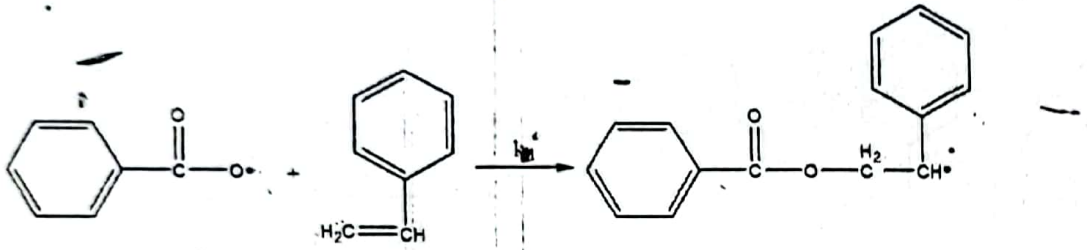
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Benzoyl peroxide * check

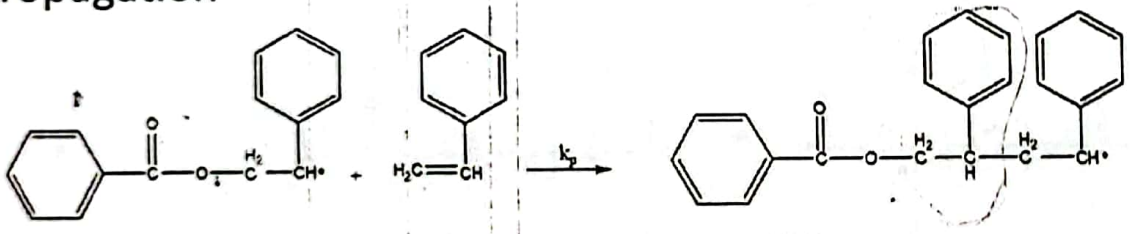
8. Explain the Initiation, propagation and termination with suitable examples for free-radical polymerization.

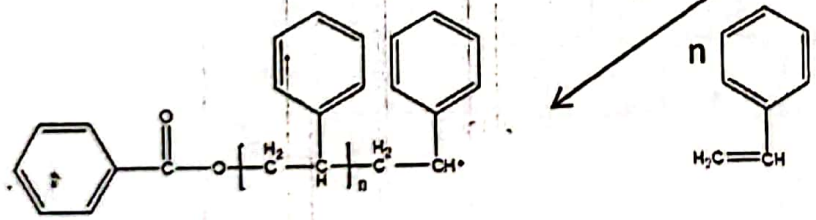
(*) 4 Initiation of polymerisation of styrene using BPO:





Propagation

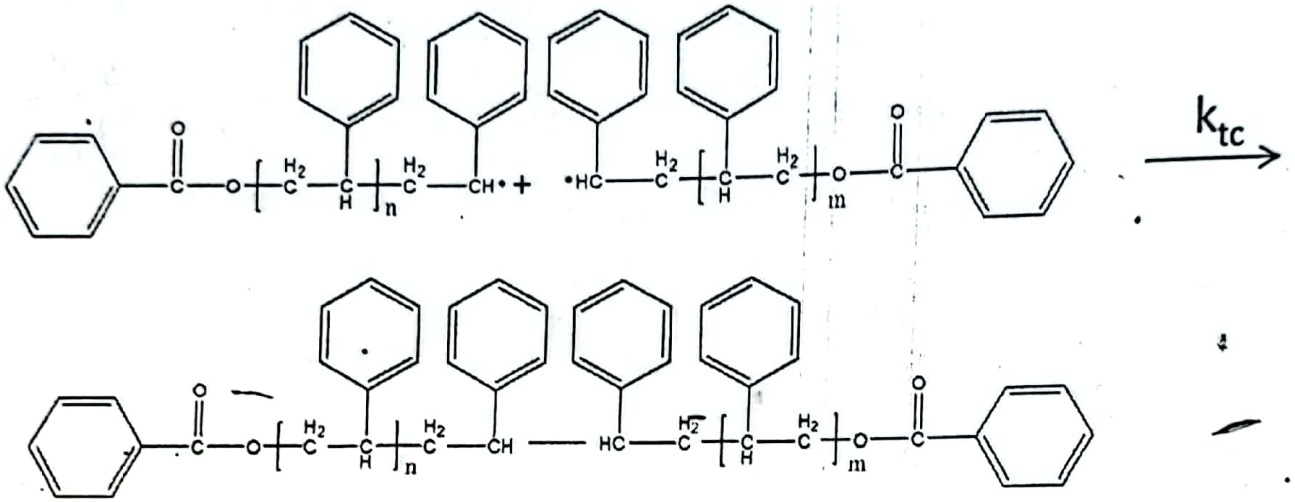




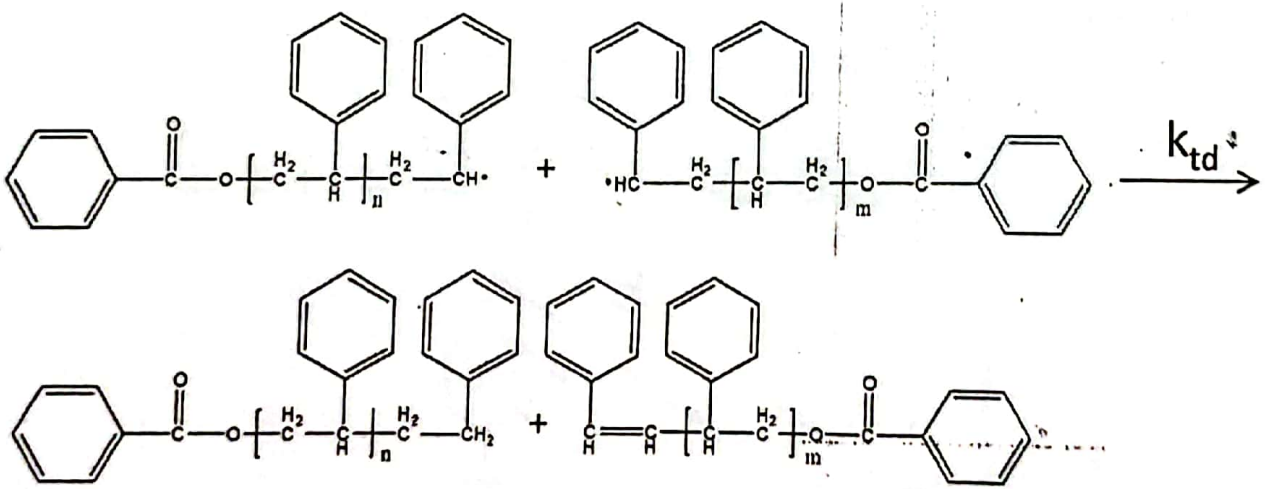


Termination

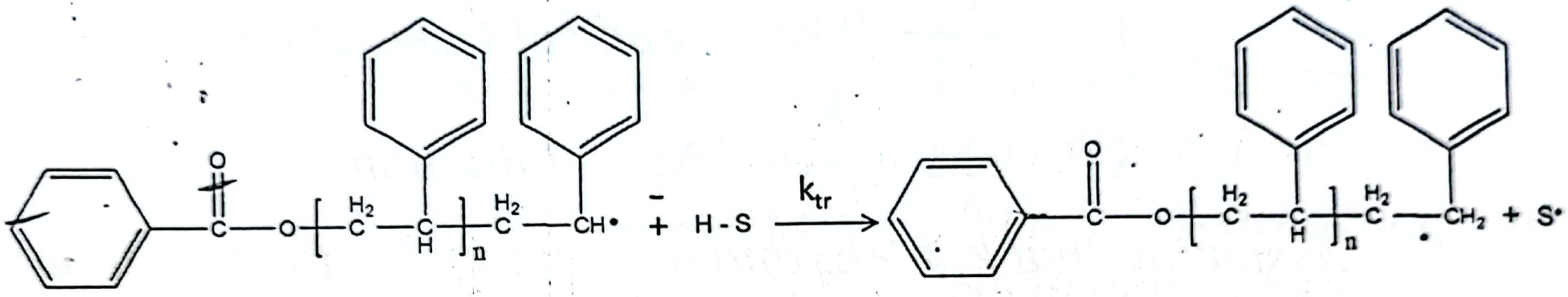
Termination by combination:



Termination by disproportionation:



Chain transfer



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Discuss the

Kinetics of Free Radical Polymerization

1. Initiation

$$I \xrightarrow{k_d} 2\,R\cdot \qquad \textit{Radical Generation}$$

$$R\cdot + M \xrightarrow{k_i} M_1\cdot \qquad \textit{Initiation}$$

Assuming that $k_i >> k_d$ and accounting for the fact that two Radicals are formed during every initiator decomposition. The rate of initiation, R_i, is given by:

$$R_i = \frac{d[M_1\cdot]}{dt} = 2fk_d[I]$$

f = efficiency of the initiator and is usually $0.3 < f > 0.8$

2. Propagation

$$M_1^{\cdot} + M \xrightarrow{k_p} M_2^{\cdot}$$

$$M_2^{\cdot} + M \xrightarrow{k_p} M_3^{\cdot}$$

$$M_3^{\cdot} + M \xrightarrow{k_p} M_4^{\cdot}$$

$$\vdots$$

$$M_x^{\cdot} + M \xrightarrow{k_p} M_{x+1}^{\cdot}$$

We assume that the reactivity of the growing chain is independent of the length of the chain.

$$R_p = -\frac{d[M]}{dt} = k_p[M^{\cdot}][M]$$

3. Termination

$$M_x\cdot + \cdot M_y \xrightarrow{k_{tc}} M_x\text{-}M_y \qquad \textit{(Combination)}$$

$$M_x\cdot + \cdot M_y \xrightarrow{k_{td}} M_x + M_y \qquad \textit{(Disproportionation)}$$

Since two radicals are consumed in every termination, then:

$$R_t = 2k_t[M\cdot]^2$$

4. Steady State Assumption

Very early in the polymerization, the concentration of radicals becomes constant because $R_i = R_t$

$$\therefore\ 2fk_d[I] = 2k_t[M\cdot]^2$$

$$2fk_d[I] = 2k_t[M\cdot]^2$$

Solving this equation for [M·]:

$$[M\cdot] = (fk_d[I]/k_t)^{1/2}$$

Substituting this into the propagation expression:

$$R_p = k_p[M\cdot][M] = k_p[M](fk_d[I]/k_t)^{1/2}$$

Since the rate of propagation, R_p, is essentially the rate of polymerization, the rate of polymerization is proportional to $[I]^{1/2}$ and [M].

5. Kinetic Chain Length, $\bar{\nu}$

Definition – *The average number of monomer units polymerized per chain initiated. This is equal to the Rate of polymerization per rate of initiation:*

$$\bar{\nu} = R_p/R_i = R_p/R_t \quad \textit{under steady state conditions.}$$

$$\bar{\nu} = \frac{k_p[M][M\cdot]}{2k_t[M\cdot]^2} = \frac{k_p[M]}{2k_t[M\cdot]}$$

$$\bar{\nu} = \frac{k_p[M]}{2(f\,k_t k_d[I])^{1/2}}$$

$\bar{\nu}$ will <u>decrease</u> with increases in initiator concentration or efficiency.

$\overline{Xn} = \bar{\nu}$ *if termination is exclusively by disproportionation.*

$\overline{Xn} = 2\bar{\nu}$ *if termination is exclusively by coupling.*

What is Chain Transfer?

Reactions in which dead polymer molecules are created, but new growing chains are initiated

- stop of the growth of molecular weight
- in a termination reaction, all free radicals are killed

$$Pn\bullet + T\text{-}A \longrightarrow Pn\text{-}T + A\bullet$$

where

T-A = chain transfer agent (monomer, initiator, solvent)
inter- and intromolecular chain transfer

T• = often hydrogen, halogen, or stable radical

A• = new radical, can start new chain

R_p not significantly affected

X_n chains are shortened, thus if there is chain transfer, we must take into account when analysing X_n.

Degree of polymerization

$$\overline{X_n} = \frac{\text{Rate of chain - growth}}{\text{Rate of termination by all processes}}$$

$$= \frac{k_p[M][M\bullet]}{k_{tc}[M\bullet]^2 + 2k_{td}[M\bullet]^2 + k_{trM}[M\bullet][M] + k_{trI}[M\bullet][I] + k_{trS}[M\bullet][S]}$$

Using steady state assumption for [M•], and rearranging:

$$\frac{1}{\overline{X_n}} = \frac{k_t R_p}{k_p^2 [M]^2} + C_M + C_I \frac{[I]}{[M]} + C_S \frac{[S]}{[M]}$$

where C_M, C_I, and C_S are transfer constants — monomer, initiator, solvent

Transfer agent can be added on purpose to tailor-make a certain $\overline{X_n}$

chain transfer coefficient.

Kinetics of Free Radical Polymerization

Kinetics of free radical polymerization can be expressed as follows.

The rate of initiation $R_i = 2fk_d[I]$

The rate of propagation $R_p = k_p[M][Pn\bullet]$

The rate of termination $R_t = 2k_t[Pn\bullet]^2$

where $k_t = k_{tc} + k_{td}$

Under the steady state assumption, the rate of initiation and the termination is equal. Thus,

$$R_i = R_t$$

From the above equations,

$$2fk_d[I] = 2k_t[Pn\bullet]^2$$

Thus, the polymer radical concentration is,

$$[Pn\bullet] = \left(\frac{fk_d[I]}{k_t}\right)^{\frac{1}{2}}$$

Inserting equation [19] into equation [16] yields,

$$R_p = k_p \left(\frac{f k_d [I]}{k_t} \right)^{\frac{1}{2}} [I]^{\frac{1}{2}} [M] = K [I]^{\frac{1}{2}} [M]$$

where K is constant.

The kinetic chain length, ν, is defined as the ratio of the rate of propagation against the rate of initiation.

$$\nu = \frac{R_p}{R_i} = \frac{k_p [M]}{2 (f k_d k_t)^{\frac{1}{2}} [I]^{\frac{1}{2}}}$$

Therefore, the number average molecular weight of polymers by free radical polymerization is,

$\overline{X_n} = \nu$ termination by disproportionation

$\overline{X_n} = 2\nu$ termination by combination

Thermodynamics of Free Radical Polymerization

The Gibbs free energy of polymerization,

$$\Delta G_p = \Delta H_p - T\Delta S_p \text{ ---------} \quad (1)$$

where ΔH_p is called the heat of polymerization defined as

$$\Delta H_p = E_{dp} - E_p \text{ ----------} \quad (2)$$

where E_{dp} and E_p are activation energy of depolymerization and propagation.

ΔG_p at low or moderate temperature is negative. As temperature increases ΔG_p become less negative. At some temperature, called celling temperature (T_c) the polymerization reaches equilibrium. In other words rate of polymerization and depolymerization become equal and

$$\Delta G_p = 0$$

The celling temperature is then obtained from eq.1 as

$$T_c = \Delta H_p / \Delta S_p$$

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Cationic Polymerization mechanism

I. Initiation

Creation of (+) active centers: Monomer + electrophiles, R+.

a) Protonic acids, H_2SO_4 + perchloric acid, $HClO_4$ involve addition of H^+ to monomer

b) Lewis acids + co-catalyst such as water, acid, organic halide

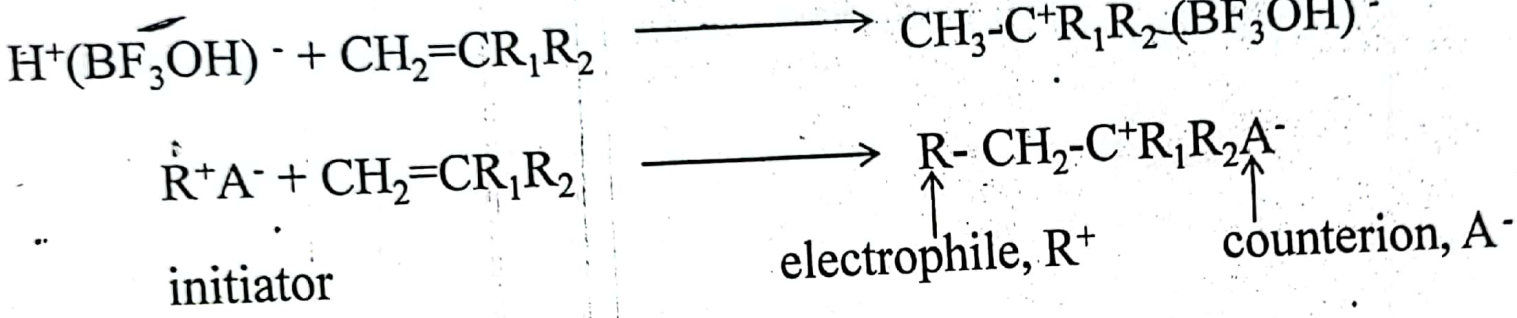
Boron trifluoride, BF_3
Aluminum chloride, $AlCl_3$
Tin tetrachloride, $SnCl_4$

First step

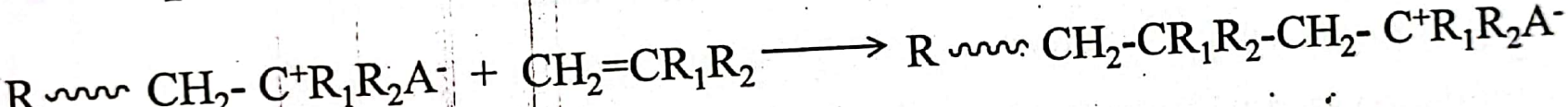
$$BF_3 + H_2O \text{ (trace)} \rightleftharpoons H^+ (BF_3OH)^-$$

$$AlCl_3 + RCl \rightleftharpoons R+ (AlCl_4)^-$$

Second Step



II. Propagation



III. Termination

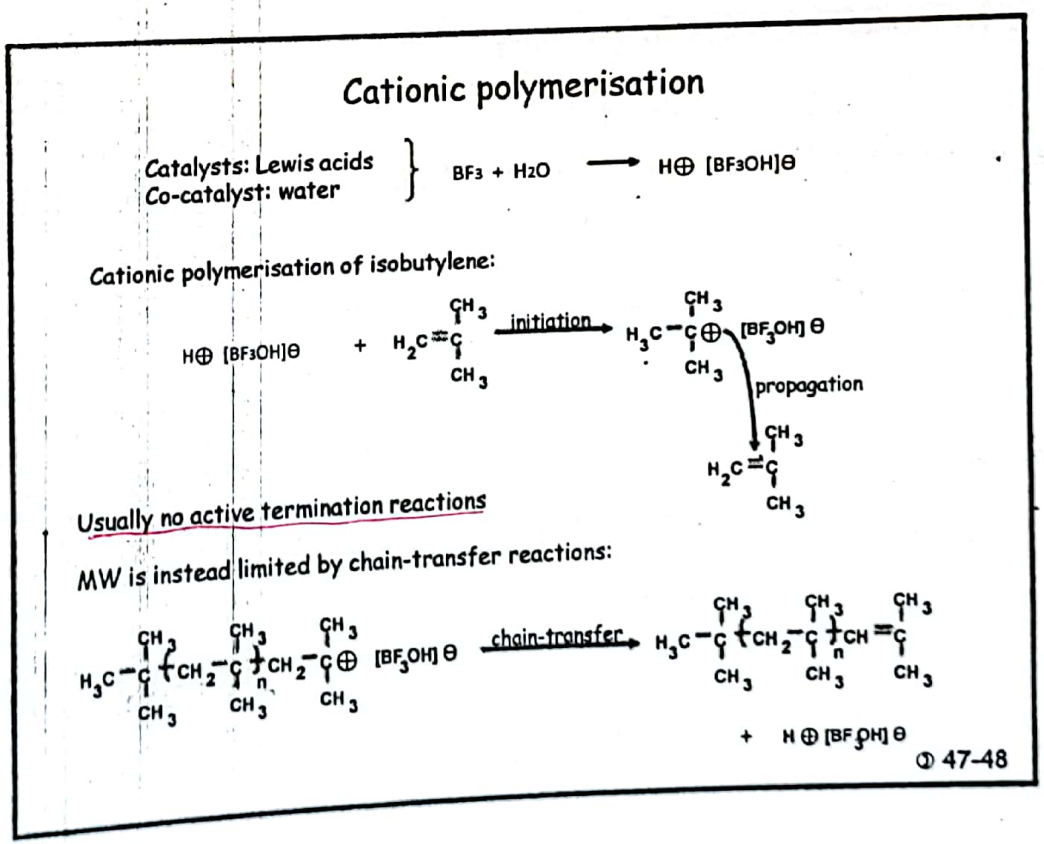
a) Unimolecular rearrangement of ion pair:



b) Chain transfer to monomer is usual

$$\sim\!\sim CH_2\text{-}C^+R_1R_2A^- + CH_2{=}CR_1R_2 \longrightarrow \underset{\text{dead polymer}}{CH_2{=}CR_1R_2} + \underset{\text{new chain}}{CH_3\text{-}C^+R_1R_2A^-}$$

chain transfer to solvent, impurities, H_2O, polymer (branching) are all common.

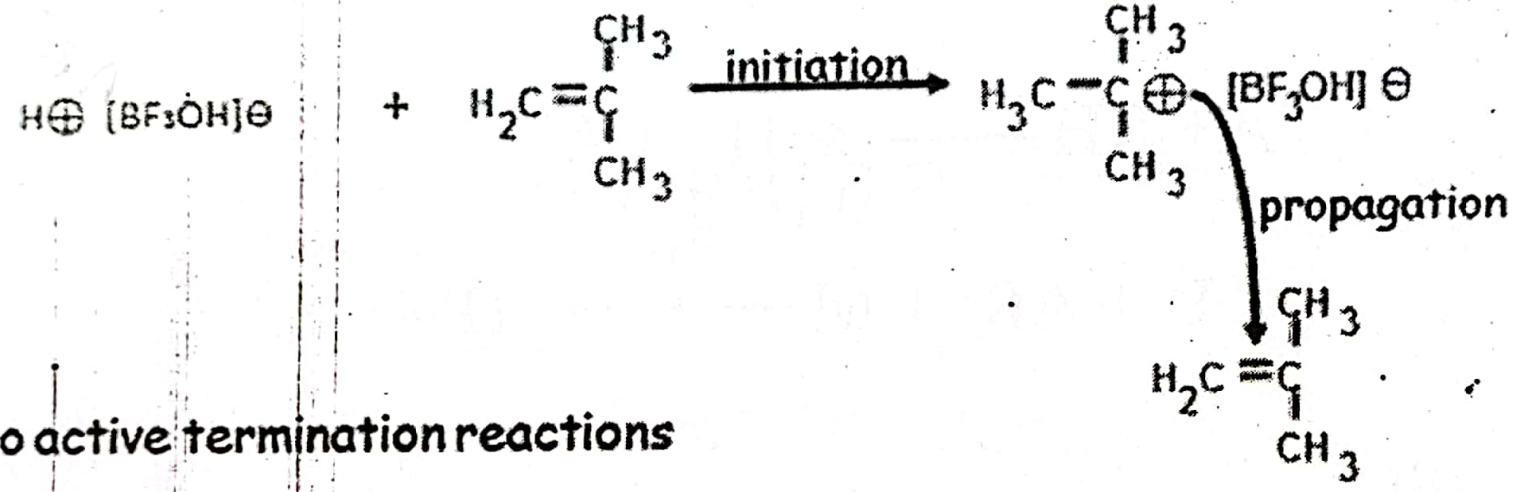


Cationic polymerisation

Catalysts: Lewis acids
Co-catalyst: water

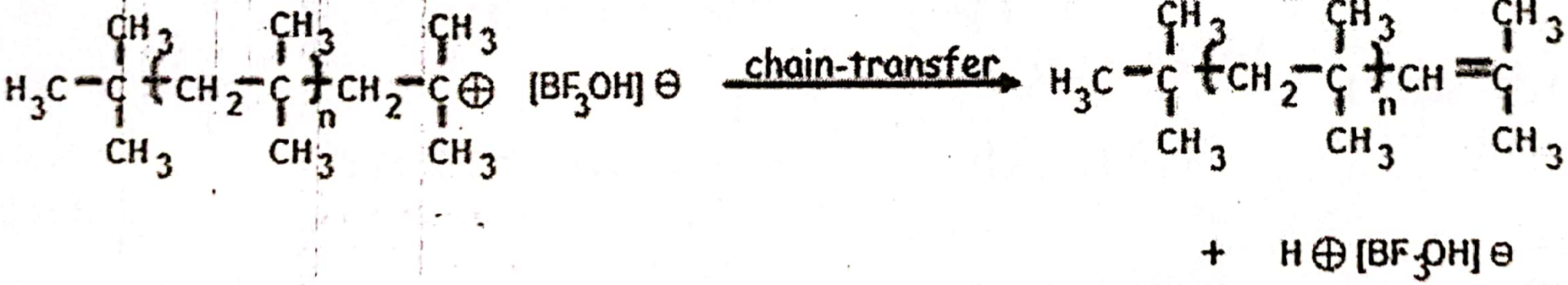
$$BF_3 + H_2O \longrightarrow H^{\oplus}\,[BF_3OH]^{\ominus}$$

Cationic polymerisation of isobutylene:



Usually no active termination reactions

MW is instead limited by chain-transfer reactions:



Kinetics of Cationic Polymerization

Most of the cationic polymerization proceed so rapidly that it is difficult to establish the steady state. So it is difficult to establish kinetics of cationic polymerization. However the following scheme seems to be valid.

Let, A be the catalyst and RH co-catalyst.

Initiation

$$A + RH \xrightarrow{K} H^+ + AR^-$$

$$H^+ + AR^- + M \xrightarrow{k_i} HM^+ + AR^-$$

Propagation

$$AR^- + HM^+ + (n-1)M \xrightarrow{k_p} HM_n^+AR^-$$

Termination

$$HM_n^+AR^- \xrightarrow{k_t} M_n + H^+AR^-$$

$$HM_n^+AR^- + M \xrightarrow{k_{tr}} M_n + HM^+AR^- \text{ (transfer)}$$

Rate of initiation

$$R_i = Kk_i[A][RH][M]$$, where [A]= catalyst concentration.

Rate of termination

$$R_t = k_t[M^+]$$, where $M^+ = HM^+AR^-$

Assuming steady state approximations, we may write , $R_i = R_t$ and from the relation,

$$[M^+] = \frac{Kk_i}{k_t}[A]\,[RH]\,[M]$$

The overall rate of polymerization R_p is given by the expression,

$$R_p = k_p[M][M^+] = K\frac{k_i k_p}{k_t}[A]\,[RH]\,[M]$$

If termination predominates over transfer, then,

$$\overline{X} = \frac{R_p}{R_t} = \frac{kp[M]\,[M^+]}{k_t[M^+]} = \frac{k_p}{k_t}[M]$$

But when transfer predominates

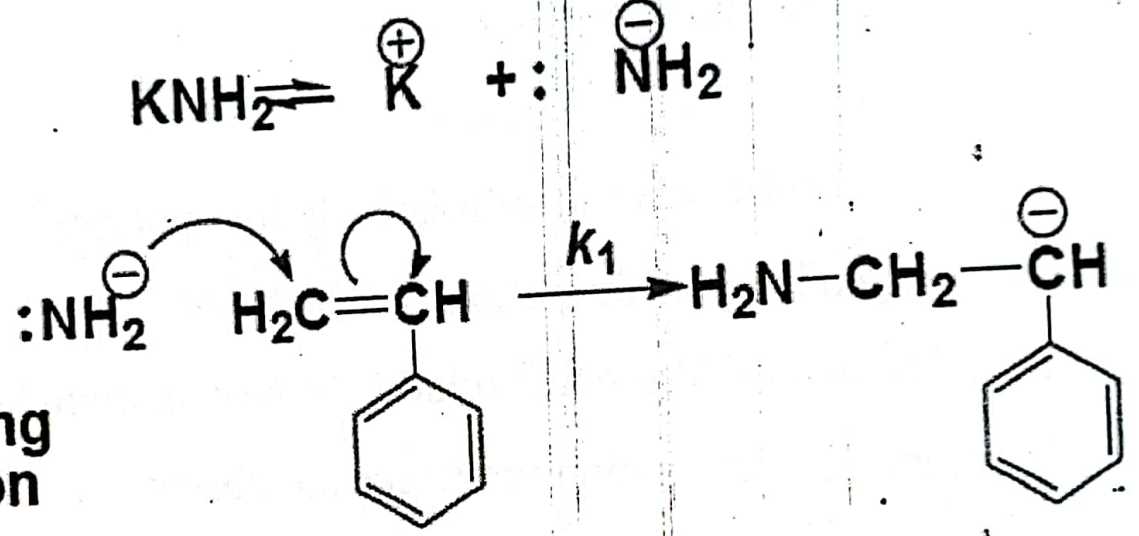
$$\overline{X} = \frac{R_p}{R_t} = \frac{k_p[M]\,[M^+]}{k_{tr}[M][M^+]} = \frac{k_p}{k_{tr}}$$

Anionic Polymerization

- Active center is (-) ion with associated (+) counterion
- Suitable monomers contain electron withdrawing groups or delocalize (-) charge
- Suitable anions are extremely strong bases
- Requires aprotic solvents of relatively low polarity
- Encourages ion pair formation
- Initiation is a rapid process relative to propagation, leading to narrow dispersity
- Spontaneous termination processes are absent

Anionic Polymerization of Styrene

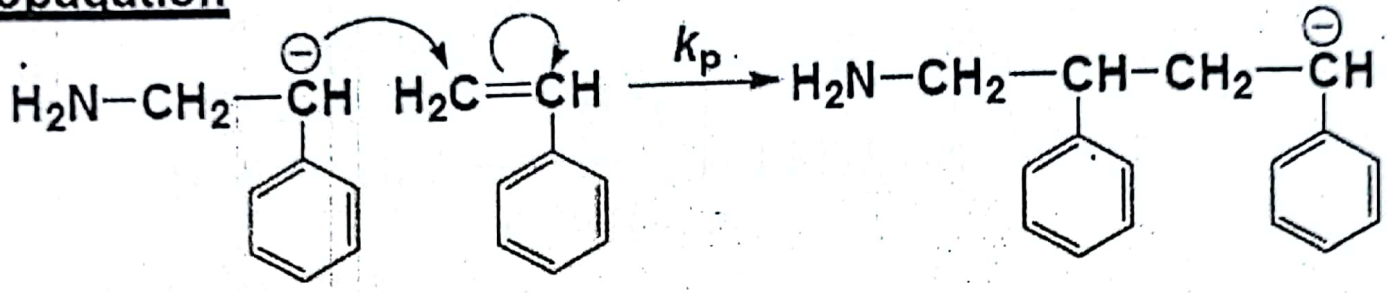
(i) **Initiation**



Rate determining step in initiation

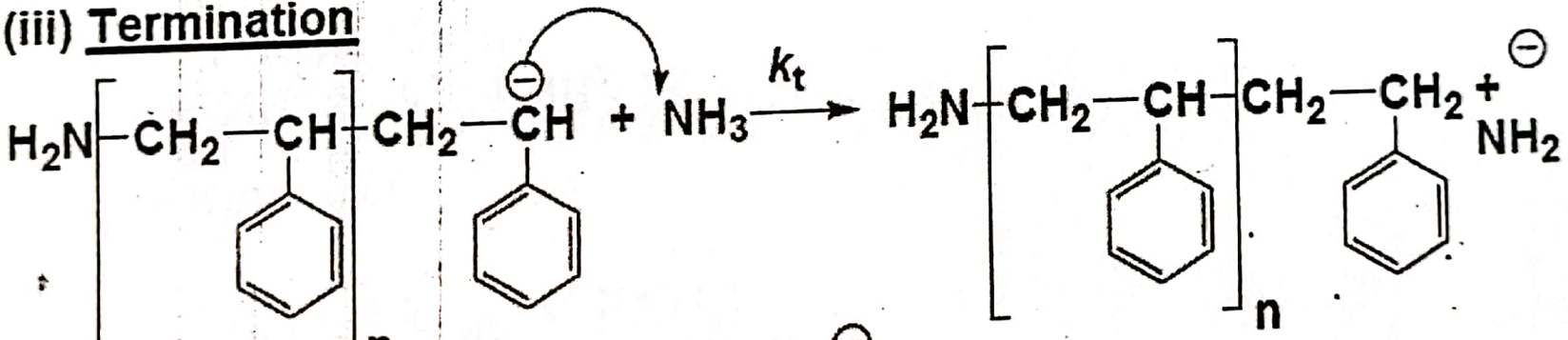
Rate $v_1 = k_1[NH_2^{\ominus}][M]$

(ii) **Propagation**



Rate $v_p = k_p[M^{\ominus}][M]$

(iii) **Termination**



Rate $v_t = k_t[M^{\ominus}][NH_3]$

2016

Kinetics of Anionic Polymerization

Initiation

$$NaNH_2 \overset{K}{\rightleftharpoons} Na^+ + NH_2^-$$

$$NH_2^- + M \xrightarrow{k_i} NH_2M^-$$

Propagation

$$NH_2M_i^- + M \xrightarrow{k_p} NH_2M^-_{i+1}$$

Termination

i) transfer to solvent

$$NH_2M^-_i + NH_3 \xrightarrow{k_{tr},NH_3} NH_2M_iH + NH_2^-$$

ii) transfer to water

$$NH_2M^-_i + H_2O \xrightarrow{k_{tr},\ H_2O} NH_2M_iH + HO^-$$

.. From equilibrium

$$K = \frac{[NH_2^-]\,[Na^+]}{[NaNH_2]} \quad ; \quad [NH^-_2] = \frac{K[NaNH_2]}{[Na^+]}$$

Rate of initiation

$$R_i = k_i\,[NH_2^-]\,[M] = \frac{k_i\,K\,[M][NaNH_2]}{[Na^+]}$$

Rate of propagation

$$R_p = k_p[M^-][M]$$

Rate of termination

$$R_{tr,NH3} = k_{tr,NH3}[M^-][NH_3]$$

$$R_{tr,H2O} = k_{tr,\,H2O}[M^-][H_2O]$$

Assuming steady state approximations, we may write ; $R_i = R_{tr, H2O}$

$$\frac{k_i K [M][NaNH_2]}{[Na^+]} = k_{tr, H2O}[M^-][H_2O]$$

$$[M^-] = \frac{k_i K [M][NaNH_2]}{[Na^+]\, k_{tr, H2O}\, [H_2O]}$$

$$R_p = k_p[M^-][M] = \frac{K k_i k_p [M]^2 [NaNH_2]}{k_{tr, H2O}\, [H_2O]\, [Na^+]}$$

$$\overline{X_n} = \frac{R_p}{R_{tr}} = \frac{k_p[M^-][M]}{k_{tr, H2O}[M^-][H_2O]} = \frac{k_p[M]}{k_{tr, H2O}[H_2O]}$$

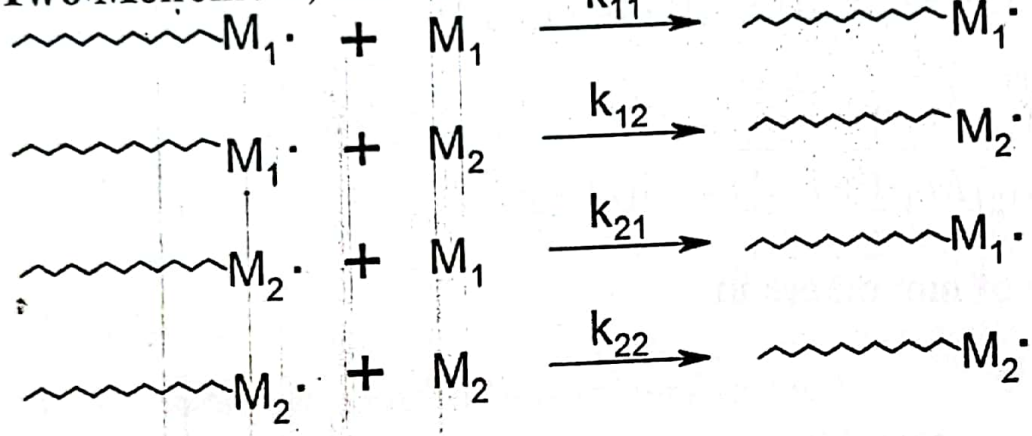
$$\frac{1}{\overline{X_n}} = \frac{C_{NH3}\,[NH_3]}{[M]} + \frac{C_{H2O}\,[H_2O]}{[M]}$$

Free - Radical Copolymerization

Four different propagation steps are possible in free - Radical Copolymerization

Only Binary Case

Two Monomers; M1 + M2



Steady State Assumption

$$-\frac{d[M_1\cdot]}{dt} = -\frac{d[M_2\cdot]}{dt} = 0$$

$$k_{12}[M_1\cdot][M_2] = k_{21}[M_2\cdot][M_1]$$

$$\frac{[M_1\cdot]}{[M_2\cdot]} = \frac{k_{21}[M_1]}{k_{12}[M_2]} \quad \text{.....①}$$

Copolymer Equation

$$-\frac{d[M_1]}{dt} = k_{11}[M_1\cdot][M_1] + k_{21}[M_2\cdot][M_1]$$

$$-\frac{d[M_2]}{dt} = k_{12}[M_1\cdot][M_2] + k_{22}[M_2\cdot][M_2]$$

$$\frac{d[M_{1\cdot}]}{d[M_2]} = \frac{[M_1]}{[M_2]}\frac{k_{11}[M_1\cdot] + k_{21}[M_2\cdot][M_1]}{k_{12}[M_1\cdot] + k_{21}[M_2\cdot][M_2]} \quad \text{.....②}$$

↑ **Instantaneous ratio of monomers in copolymer**

From ① and ②

$$\frac{d[M_1]}{d[M_2]} = \frac{1 + r_1\frac{[M_1]}{[M_2]}}{1 + r_2\frac{[M_2]}{[M_1]}}$$

← **Copolymer Eq.**

where $r_2 = \frac{k_{22}}{k_{21}}$, $r_1 = \frac{k_{11}}{k_{12}}$ ← **monomer reactivity ratio**

Meaning of r & Definition of f_1, F_1

Meaning of r

r_1 characterizes the reactivity of the 1 radical with respect to the two monomers, 1 and 2

$r_1 > 1$ then homopolymerization growth is preferred

$r_1 = 0$ then only reaction with 2 will occur

Define f_1, F_1

f_1, f_2 : mole fractions of monomers in feed

F_1, F_2 : mole fractions of monomers in polymer

$$f_1 = 1 - f_2 = \frac{[M_1]}{[M_1 + M_2]} \quad \cdots\cdots ③$$

$$F_1 = 1 - F_2 = \frac{d[M_1]}{d[M_1] + d[M_2]} \quad \cdots\cdots ④$$

From ③, ④

$$F_1 = \frac{r_1 f_1^2 + f_1 f_2}{r_1 f_1^2 + 2 f_1 f_2 + r_2 f_2^2} \quad \cdots\cdots ⑤$$

Living Polymerization

- No termination step and no chain transfer. Polymerisation stop when there is no more monomer. Once the monomer is replenish, polymerisation resumed.

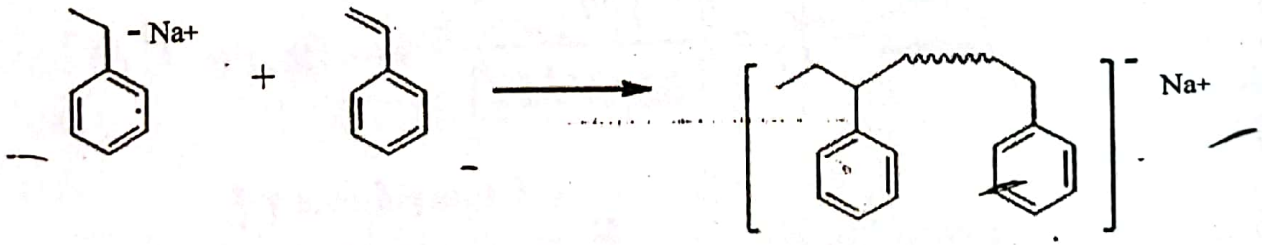
-Advantage: 1. Control polydispersity
2. Molecular weight
3. Block copolymer

Example of living radical polymerisation - anionic polymerisation
Sodium dihydronaphthylide with styrene

Initiation:



Propagation:



Comparison of Step-Reaction and Chain-Reaction Polymerization

Step Reaction	Chain Reaction
1. Growth occurs throughout matrix by reaction between monomers, oligomers, and polymers	1. Growth occurs by successive addition of monomer units to limited number of growing chains
2. $\overline{DP}$ low to moderate	2. $\overline{DP}$ can be very high Monomer
3. Monomer consumed rapidly while molecular weight increases slowly	3. consumed relatively slowly, but molecular weight increases rapidly
4. No initiator needed; same reaction mechanism throughout	4. Initiation and propagation mechanisms different
5. No termination step; end groups still reactive	5. Usually chain-terminating step involved
6. Polymerization rate decreases steadily as functional groups consumed	6. Polymerizaion rate increases initially as initiator units generated; remains relatively constant until monomer depleted

[a] $\overline{DP}$, average degree of polymerization.